Stabilizing Thompson Sampling with Null Hypothesis Bayesian Response-Adaptive Randomization

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Abstract

Response-adaptive randomization (RAR) methods can be used to adapt randomization probabilities based on accumulating data, aiming to increase the probability of allocating patients to effective treatments. A popular RAR method is Thompson sampling, which randomizes patients proportionally to the Bayesian posterior probability that each treatment is the most effective. However, its high variability early in a trial can also increase the risk of assigning patients to inferior treatments and lead to inferential problems such as confidence interval undercoverage. We propose a principled method based on Bayesian hypothesis testing and model averaging to mitigate these issues: We introduce a null hypothesis postulating equal effectiveness of treatments. Bayesian model averaging then induces shrinkage toward equal randomization probabilities, with the degree of shrinkage controlled by the prior probability of the null hypothesis. Equal randomization and Thompson sampling arise as special cases when the prior probability is set to one or zero, respectively. Simulated and real-world examples illustrate that the method balances highly variable Thompson sampling with static equal randomization. A simulation study demonstrates that the method can mitigate issues with Thompson sampling and has comparable statistical properties to Thompson sampling with common ad hoc modifications such as power transformation and probability capping. We implement the method in the open-source R package `brar`, enabling experimenters to easily perform null hypothesis Bayesian RAR and support more effective randomization of patients.

Keywords: Adaptive trials, A/B testing, Bayes factor, Bayesian model averaging, spike-and-slab prior

1 Introduction

Response-adaptive randomization (RAR) methods, sometimes also called outcome-adaptive randomization methods, randomly allocate experimental units (e.g., patients or animals) to treatments based on accumulating data (Hu and Rosenberger, 2006; Thall and Wathen, 2007; Berry et al., 2010; Grieve, 2016; Robertson et al., 2023). A popular approach is Thompson sampling (Thompson, 1933), which randomizes participants proportionally to the Bayesian posterior probability that each treatment is the most effective. Such RAR methods are attractive because they naturally balance gathering information on treatment effectiveness and assigning subjects to effective treatments.

Despite their benefits, RAR methods present various challenges. For example, if not properly adjusted for, Thompson sampling can introduce inferential problems, such as inflated type I error rates or bias in effect estimation. Furthermore, although Thompson sampling allocates participants to the more effective treatment on average, it may increase the probability of allocating participants to inferior treatments compared to equal randomization, particularly when treatment effects are small. This can undermine the method's intended objective and raise ethical concerns (Thall et al., 2015; Robertson et al., 2023). Consequently, there has been interest in modifying Thompson sampling to mitigate these issues.

One popular modification proposed by Thall and Wathen (2007) uses the randomization probability $\pi = p^c / \{p^c + (1-p)^c\}$, where p is the posterior probability that the experimental treatment is more effective than the control treatment, and c is an additional parameter that controls the variability of the method. Setting $c = 1$ produces Thompson sampling, while $c < 1$ reduces variability, with $c = 1/2$ being often recommended. Another approach to reducing extreme randomization probabilities is capping them, for instance, setting them to 10% or 90% if a method assigns more extreme probabilities (Thall and Wathen, 2007; Lee and Lee, 2021). Finally, RAR methods are often combined with "burn-in" periods at the start of the study, during which units are randomized with equal probabilities to mitigate high variability (Thall and Wathen, 2007; Wathen and Thall, 2017; Robertson et al., 2023; Sverdlov et al., 2025).

Several simulation studies have compared Thompson sampling modifications (e.g., Du et al., 2017; Viele et al., 2019, 2020; Tang et al., 2025), finding that, if appropriately specified, they can mitigate its limitations. However, such ad hoc modifications conflict with principles of coherent Bayesian learning. For example, a transformed or capped posterior probability no longer corresponds to an actual posterior probability and cannot serve as a genuine prior for future data. This raises the question of whether a RAR method can be devised that mitigates the excessive variability of Thompson sampling while remaining coherent with Bayesian principles, and if so, how it relates to these ad hoc modifications.

Principled Bayesian alternatives to Thompson sampling have also been proposed outside the clinical trials field, particularly in the multi-armed bandit literature (see, e.g., Lattimore and Szepesvári, 2020). For instance, RAR based on the Gittins index uses an allocation rule that prioritizes treatments groups with the greatest expected reward (Gittins et al., 2011). Another alternative is the Bayesian upper confidence bound method, which allocates treatments based on posterior quantiles and offers theoretical performance guarantees (Kaufmann et al., 2012). These methods were primarily developed for online decision problems, but are rarely used in clinical trials, where additional constraints such as type I error rate control or avoiding extreme randomization probabilities are central (Villar et al., 2015a,b).

In this paper, we propose a novel RAR method that reduces variability in a coherent Bayesian manner, which we term "null hypothesis Bayesian RAR". The idea is to consider a null hypothesis postulating that treatments are equally effective. This is often equivalent to using a "spike-and-slab" prior (Raftery et al., 1997) which is a mixture of a point mass at equal effectiveness and a probability density elsewhere (Spiegelhalter et al., 2004, Section 5.5.4). The prior probability of the null hypothesis determines the mixture weight. As we will show, setting this prior probability to zero produces equal randomization, whereas setting it to one produces Thompson sampling. The proposed method thus interpolates between equal randomization and Thompson sampling in a coherent Bayesian way. As a by-product, posterior probabilities and Bayes factors are also obtained. These can be used to monitor evidence of the effectiveness of each treatment in a manner that is aligned with the randomization probabilities.

In the following Section 2 we introduce the general idea of the method in more detail, followed by tailoring it to the setting of approximately normal effect estimates (Section 3) and binary outcomes (Section 4). In Section 5, we then illustrate the method on data from the ECMO trial (Bartlett et al., 1985), followed by evaluating its statistical properties in a simulation study (Section 6). The paper ends with concluding remarks on advantages, limitations, and opportunities for future research (Section 7). Appendix A illustrates our R package `brar` for performing Bayesian RAR. Appendix B provides further details on our simulation study.

2 Null hypothesis Bayesian RAR

We now explain the general idea of null hypothesis Bayesian RAR, without going into specifics such as data distribution or computation (these will follow in the subsequent sections). Throughout we will assume that we have observed data y and we want to use these to randomize a future experimental unit. We start with the basic but important setting of one control and one treatment group, and extend it afterwards to multiple treatment groups.

2.1 Two group comparisons

In case there is a control group and only one treatment group, we consider the hypotheses:

$$H_{-}: \text{Treatment is less effective than control}$$
$$H_{0}: \text{Treatment and control are equally effective}$$
$$H_{+}: \text{Treatment is more effective than control.}$$

How exactly these statements are translated into statistical hypotheses related to parameters depends on the type of data and model used, but often relates to an effect size parameter being less, equal, or greater than zero. There may also be situations where there is no control group but only two competing treatments. In this case, the method is still applicable but with the control group replaced by a reference group (the choice of the reference may be somewhat arbitrary). This hypothesis setup is similar to the RAR method from Cai et al.

(2013), though with the addition of the null hypothesis H_0, which as we will show, introduces a different behavior of the resulting randomization probabilities.

The Bayesian posterior probability of a hypothesis $H_i \in \{H_-, H_0, H_+\}$ is then

$$\Pr(H_i \mid y) = \frac{p(y \mid H_i)\Pr(H_i)}{\sum_{j\in\{-,0,+\}} p(y \mid H_j)\Pr(H_j)} = \left\{\sum_{j\in\{-,0,+\}} \mathrm{BF}_{ji}(y)\frac{\Pr(H_j)}{\Pr(H_i)}\right\}^{-1} \tag{1}$$

where $\Pr(H_i)$ is the prior probability of hypothesis H_i, $p(y \mid H_i) = \int_\Theta p(y \mid \theta)\, p(\theta \mid H_i)\, \mathrm{d}\theta$ is the marginal likelihood of the data y under H_i obtained from marginalizing the likelihood $p(y \mid \theta)$ with respect to the prior distribution $p(\theta \mid H_i)$ assigned to the model parameters $\theta \in \Theta$ under H_i, and

$$\mathrm{BF}_{ji}(y) = \frac{\Pr(H_j \mid y)}{\Pr(H_i \mid y)} \bigg/ \frac{\Pr(H_j)}{\Pr(H_i)} = \frac{p(y \mid H_j)}{p(y \mid H_i)} \tag{2}$$

is the Bayes factor contrasting H_j to H_i (Jeffreys, 1939; Good, 1958). The Bayes factor (2) is the updating factor of the prior odds of H_j to H_i to the corresponding posterior odds (first equality), which is equivalent to the ratio of marginal likelihoods of the data under H_j and H_i (second equality). The posterior probabilities (1) can thus be computed from the marginal likelihoods of the data under each considered hypotheses along with their prior probabilities, or from the set of Bayes factors and prior hypothesis odds relative to some reference hypothesis (Kass and Raftery, 1995).

Regardless in which way they are computed, the question is how to translate posterior probabilities into randomization probabilities. We propose to randomize a future unit to the treatment group with probability

$$\pi = \Pr(H_+ \mid y) + \Pr(H_0 \mid y) \times \frac{1}{2}, \tag{3}$$

while the probability to randomize to the control group is consequently

$$1 - \pi = \Pr(H_- \mid y) + \Pr(H_0 \mid y) \times \frac{1}{2}. \tag{4}$$

Figure 1 shows such randomization probabilities for different combinations of posterior probabilities. We can see that the randomization probability π shrinks towards 50% as the posterior probability of the null hypothesis $\Pr(H_0 \mid y)$ increases. For example, if $\Pr(H_+ \mid y) = 0.2$, $\Pr(H_- \mid y) = 0.3$, and $\Pr(H_0 \mid y) = 0.5$, as indicated by the black asterisk in Figure 1, the probability to randomize to the treatment group is $\pi = 0.2 + 0.5/2 = 45\%$.

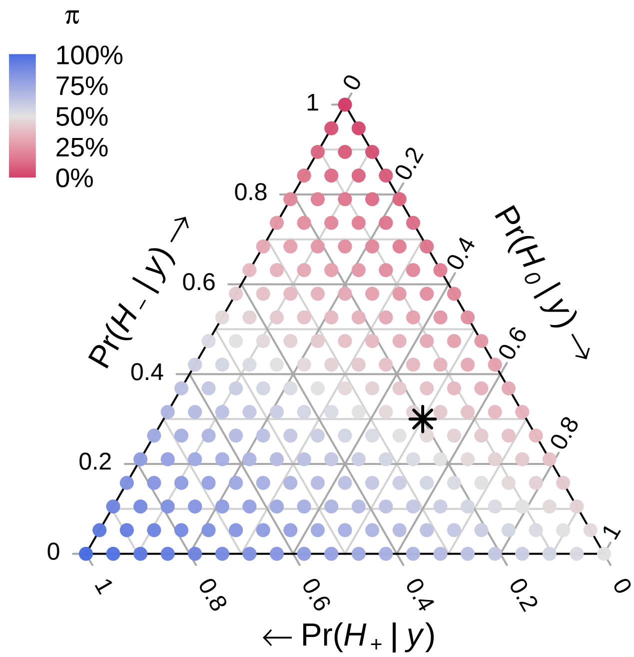

Figure 1: Ternary plot where the color of each point depicts the treatment randomization probability π for a particular combination of posterior probabilities of H_+ (bottom axis), H_- (left axis), and H_0 (right axis). For example, the black asterisk denotes the combination where $\Pr(H_+ \mid y) = 0.2$, $\Pr(H_- \mid y) = 0.3$, and $\Pr(H_0 \mid y) = 0.5$ with corresponding randomization probability $\pi = 45\%$.

This scheme can be motivated as Bayesian hypothesis averaged randomization probability where the hypothesis-specific treatment randomization probabilities are 0%, 50%, and 100%, under H_-, H_0, and H_+, respectively. That is, if we would know that H_+ or H_- is true, we should assign the next patient to the treatment or control group, respectively, with probability one to maximize utility (patient benefit). On the other hand, if we knew that H_0 is true, randomizing the assignment with $\pi = 50\%$ seems sensible.

Interestingly, the randomization scheme (3) reduces to Thompson sampling when the prior probability of H_0 is set to zero since then $\Pr(H_0 \mid y) = 0$ regardless of the data, and consequently $\pi = \Pr(H_+ \mid y)$. However, the scheme induces shrinkage toward randomization probabilities of $\pi = 50\%$ otherwise. In the most extreme case when $\Pr(H_0) = 1$, equal randomization is obtained as then $\Pr(H_0 \mid y) = 1$ regardless of the data, and consequently $\pi = 50\%$. The scheme thus interpolates between Thompson sampling and equal randomization in a coherent Bayesian way.

2.2 More than two groups

Suppose there are $K > 1$ treatment groups in addition to the control group. In this case, we may modify the procedure and consider the hypotheses:

H_-: All treatments are less effective than control

H_0: All treatments are equally effective as control

H_{+i}: Treatment i is more effective than control and all other treatments

for $i = 1, \ldots, K$. Posterior probabilities of each hypothesis can be computed from the marginal likelihoods and prior hypotheses probabilities with the summation in (1) extended to encompass all hypotheses (i.e., summing over $j \in \{-, 0, +1, \ldots, +K\}$). Similarly, they can be translated into randomization probabilities

$$\pi_i = \Pr(H_{+i} \mid y) + \Pr(H_0 \mid y) \times \frac{1}{K+1} \tag{5}$$

with corresponding control randomization probability

$$1 - \sum_{i=1}^{K} \pi_i = \Pr(H_- \mid y) + \Pr(H_0 \mid y) \times \frac{1}{K+1}, \tag{6}$$

which reduce to the randomization probabilities (3) and (4) for $K = 1$.

Also in the multi-treatment case, the randomization probabilities are shrunken towards equal randomization $\pi = 1/(K+1)$ by introducing the null hypothesis H_0. Similarly, Thompson sampling and equal randomization are obtained by setting $\Pr(H_0) = 0$ and $\Pr(H_0) = 1$, respectively, since then the posteriors $\Pr(H_0 \mid y) = 0$ and $\Pr(H_0 \mid y) = 1$ are obtained for any observed data y.

In this paper, we focus on schemes (5) and (6), which induce shrinkage toward equal randomization. However, it is important to note that there are alternatives to this approach. For example, it may be desired to shrink to different "baseline" randomization probabilities than equal randomization probabilities. This can be achieved by modifying the multiplicative factor of $\Pr(H_0 \mid y)$ in (5) from $1/(K+1)$ to the desired baseline randomization probability. For example, the aim may be to test all K pairwise differences between the treatments and the common control using Dunnett's test (Dunnett, 1955). A $\sqrt{K} : 1 : \cdots : 1$ square-root allocation of the control to treatments is commonly used to maximize the power of the test (Kieser, 2020, p.156). We may therefore use

$$\pi_i = \Pr(H_{+i} \mid y) + \Pr(H_0 \mid y) \times \frac{1}{K + \sqrt{K}}$$

which leads to the control randomization probability

$$1 - \sum_{i=1}^{K} \pi_i = \Pr(H_- \mid y) + \Pr(H_0 \mid y) \times \frac{\sqrt{K}}{K + \sqrt{K}}.$$

These correctly shrink RAR probabilitites towards the Dunnett-type randomization probabilities, $\pi_i = 1/(K + \sqrt{K})$ for treatment $i = 1, \ldots, K$, and $1 - \sum_{i=1}^{K} \pi_i = \sqrt{K}/(K + \sqrt{K})$ for control.

2.3 Asymptotic randomization probabilities

It is of interest to understand how RAR probabilities behave as more data are collected. Under ordinary non-RAR based sampling, it is well known that the posterior distribution is "consistent", meaning that as more data are generated under a hypothesis the corresponding posterior hypothesis probability tends to one (Dawid, 2011). In Section 3.1 we provide empirical support that posterior consistency also holds for the RAR setting. This means that if a treatment (or the control) is superior, the corresponding posterior and randomization probabilities will tend to one. Conversely, if all treatments and control are equally effective, the posterior probability of H_0 will approach one, and the randomization probabilities will hence approach the specified baseline randomization probabilities (e.g., equal or Dunnett-type). This is an important difference to standard Thompson sampling, whose randomization probability does not converge to the baseline randomization probability but remains random (Zhang et al., 2020; Kalvit and Zeevi, 2021; Deliu and Villar, 2025).

3 Null hypothesis Bayesian RAR under normality

We will expand on null hypothesis Bayesian RAR in the setting where the data are summarized by an asymptotically normally distributed effect estimate. This does not mean that the raw data (e.g., a vector of outcomes and a matrix of covariates), from which the estimate is computed, need to be normally distributed. For instance, the regression coefficients from generalized linear models estimated using maximum likelihood satisfy asymptotic normality even if the data are themselves not normally distributed. This is useful because it allows us to efficiently compute posterior and randomization probabilities without the need for simulation. Although this framework is widely applicable, improvements may be possible by considering the exact data distribution. Section 4 will demonstrate this for binary data.

3.1 Two group comparisons

Assume that the data are summarized by $y = \{\hat{\theta}, \sigma\}$, where $\hat{\theta}$ is an effect estimate of the true treatment effect θ that quantifies the effect of a treatment over the control and σ is the standard error of the estimate. For example, $\hat{\theta}$ could be an estimated (standardized) mean difference, log odds/rate/hazard ratio, risk difference, or regression coefficient. The standard error is often of the form $\sigma = \lambda/\sqrt{n}$, where n is the effective sample size and λ the standard deviation of one effective observation. Suppose further that the estimate is (at least approximately) normally distributed around θ with its variance σ^2 assumed to be known, i.e., $\hat{\theta} \mid \theta \sim \mathrm{N}(\theta, \sigma^2)$.

If the effect θ is oriented such that a positive effect indicates treatment benefit, the three hypotheses from Section 2.1 translate into:

$$H_-\colon \theta < 0 \qquad \text{vs.} \qquad H_0\colon \theta = 0 \qquad \text{vs.} \qquad H_+\colon \theta > 0.$$

The null hypothesis H_0 is a simple hypothesis with no free parameters, or equivalently, a point (Dirac) prior at zero. The H_- and H_+ hypotheses are composite hypotheses and require prior distributions for the effect θ. A natural choice is a $\theta \sim \mathrm{N}(\mu, \tau^2)$ normal distribution whose support is truncated to the negative and positive side, respectively. This distribution may be specified based on prior knowledge (e.g., previous studies). In the absence of prior knowledge, it is sensible to center the prior on zero ($\mu = 0$) to represent clinical equipoise (Freedman, 1987), as a zero-centered prior gives equal probability to harmful and beneficial effects. Due to H_0 being a point hypothesis, the prior of θ under H_- and H_+ must be proper ($\tau < \infty$), as otherwise the posterior probability of H_0 is always 1 (the Jeffreys-Lindley paradox, see, e.g., Wagenmakers and Ly, 2022). The value of the prior standard deviation τ should therefore be chosen so that the prior is not too vague, for example, so that it spans realistic values of the effect θ. Another sensible default choice is a "unit-information prior" which sets τ to the standard deviation of one observation (Kass and Wasserman, 1995).

To compute posterior hypothesis probabilities, specification of prior hypothesis probabilities is required. The prior probability of the null hypothesis $\Pr(H_0)$ represents the a priori plausibility of an absent effect, and also acts as a tuning parameter that controls the degree of variability of RAR. An intuitive default is $\Pr(H_0) = 0.5$ as it represents the equipoise position of equal probability of an absent effect relative to a present (either harmful or beneficial) effect (Johnson, 2013). For a given $\Pr(H_0)$, it is then natural to set the prior probabilities of the other two hypotheses to $\Pr(H_+) = \{1 - \Pr(H_0)\} \times \Phi(\mu/\tau)$ and $\Pr(H_-) = \{1 - \Pr(H_0)\} \times \Phi(-\mu/\tau)$ so that to the prior distribution averaged over H_- and H_+ is again the $\mathrm{N}(\mu, \tau^2)$ normal distribution that was truncated in the first place. For example, when setting $\Pr(H_0) = 0.5$ and specifying a zero-centered prior ($\mu = 0$) as in Figure 2, we obtain $\Pr(H_+) = \Pr(H_-) = 0.5 \times 0.5 = 0.25$. Averaging the prior over the three

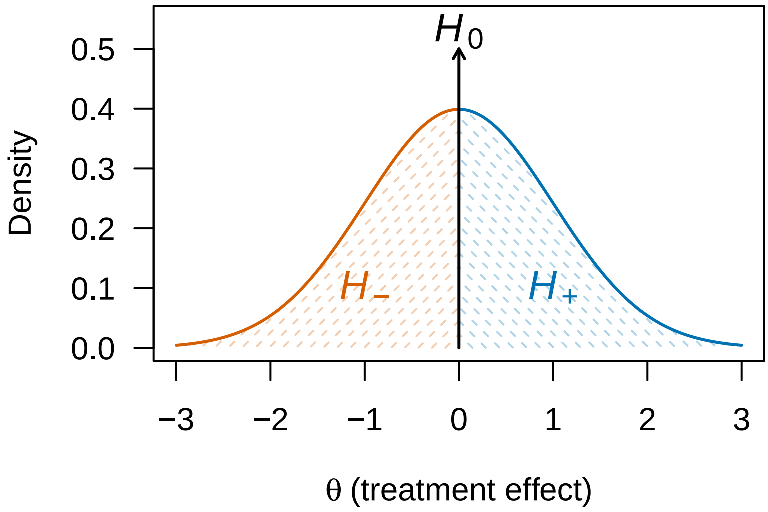

Figure 2: Illustration of spike-and-slab prior for the effect θ. A point prior at 0 is assumed under H_0. A normal prior $\theta \sim \mathrm{N}(0, 1)$ with support truncated to the positive or negative side is assumed under H_+ and H_-, respectively. These priors are averaged assuming prior hypothesis probabilities $\Pr(H_0) = 0.5$, $\Pr(H_+) = 0.25$, and $\Pr(H_-) = 0.25$.

hypotheses leads then to a spike-and-slab prior, as illustrated in Figure 2.

With the prior densities and prior hypothesis probabilities specified, we can compute the marginal likelihood of the observed effect estimate under each hypothesis, and in turn obtain posterior probabilities (1). In the conjugate normal likelihood and prior framework, all of them can be straightforwardly derived in closed-form. Denoting by $\mathrm{N}(x \mid m, v)$ the normal density function with mean m and variance v evaluated at x, the marginal likelihoods are given by

$$p(\hat{\theta} \mid H_-) = \mathrm{N}(\hat{\theta} \mid \mu, \sigma^2 + \tau^2) \times \frac{\Phi(-\mu_*/\tau_*)}{\Phi(-\mu/\tau)} \tag{7a}$$

$$p(\hat{\theta} \mid H_0) = \mathrm{N}(\hat{\theta} \mid 0, \sigma^2) \tag{7b}$$

$$p(\hat{\theta} \mid H_+) = \mathrm{N}(\hat{\theta} \mid \mu, \sigma^2 + \tau^2) \times \frac{\Phi(\mu_*/\tau_*)}{\Phi(\mu/\tau)} \tag{7c}$$

with posterior mean and variance

$$\mu_* = \frac{\hat{\theta}/\sigma^2 + \mu/\tau^2}{1/\sigma^2 + 1/\tau^2} \qquad \text{and} \qquad \tau_*^2 = \frac{1}{1/\sigma^2 + 1/\tau^2}.$$

Taking ratios of marginal likelihoods produces the Bayes factors

$$\mathrm{BF}_{+0}(\hat{\theta}) = \exp\left[-\frac{1}{2}\left\{\frac{(\hat{\theta}-\mu)^2}{\sigma^2+\tau^2} - \frac{\hat{\theta}^2}{\sigma^2}\right\}\right] \times \frac{\Phi(\mu_*/\tau_*)}{\Phi(\mu/\tau)} \Bigg/ \sqrt{1+\frac{\tau^2}{\sigma^2}}$$

$$\mathrm{BF}_{+-}(\hat{\theta}) = \frac{\Phi(\mu_*/\tau_*)}{\Phi(\mu/\tau)} \Bigg/ \frac{\Phi(-\mu_*/\tau_*)}{\Phi(-\mu/\tau)},$$

and the Bayes factors for other hypothesis comparisons can be obtained by transitivity and reciprocity, for example, $\mathrm{BF}_{-0} = \mathrm{BF}_{+0} / \mathrm{BF}_{+-}$. Posterior probabilities can now be obtained by plugging the Bayes factors and prior odds into (1). These Bayes factors and posterior probabilities can be monitored as data accumulate to see how the evidence for the hypotheses changes. Moreover, they can be plugged into (3) to obtain the randomization probability

$$\pi = \frac{\overbrace{\Phi\left(\mu_*/\tau_*\right)}^{\text{Thompson Sampling}}}{1+\frac{\Pr(H_0)}{1-\Pr(H_0)} \times \mathrm{BF}_{01}(\hat{\theta})} + \frac{\overbrace{1/2}^{\text{Equal Randomization}}}{1+\frac{1-\Pr(H_0)}{\Pr(H_0)} / \mathrm{BF}_{01}(\hat{\theta})} \tag{8}$$

where

$$\mathrm{BF}_{01}(\hat{\theta}) = \exp\left[-\frac{1}{2}\left\{\frac{\hat{\theta}^2}{\sigma^2} - \frac{(\hat{\theta}-\mu)^2}{\sigma^2+\tau^2}\right\}\right] \times \sqrt{1+\frac{\tau^2}{\sigma^2}}$$

is the Bayes factor contrasting H_0 to the alternative $H_1 = H_- \cup H_+$ with normal prior $\theta \mid H_1 \sim \mathrm{N}(\mu, \tau^2)$. As expected, the randomization probability (8) approaches equal randomization as the prior probability of H_0 increases to one (i.e., $\pi \to 50\%$ as $\Pr(H_0) \nearrow 1$), whereas it approaches the ordinary Bayesian posterior tail probability of $\theta > 0$ based on a normal prior $\theta \sim \mathrm{N}(\mu, \tau^2)$ as the prior probability of H_0 decreases to zero (i.e., $\pi \to 1 - \Phi\{(0-\mu_*)/\tau_*\} = \Phi(\mu_*/\tau_*)$ as $\Pr(H_0) \searrow 0$). Similarly, the randomization probability shrinks towards 50% as the data provide more evidence for H_0 ($\pi \to 50\%$ as $\mathrm{BF}_{01} \nearrow \infty$), whereas it approaches the ordinary posterior tail probability as the evidence for H_1 increases ($\pi \to \Phi(\mu_*/\tau_*)$ as $\mathrm{BF}_{01} \searrow 0$). In contrast to ad hoc modifications of Thompson sampling (e.g., capping or power transformations), shrinkage of the posterior tail probability thus depends on the accumulated data.

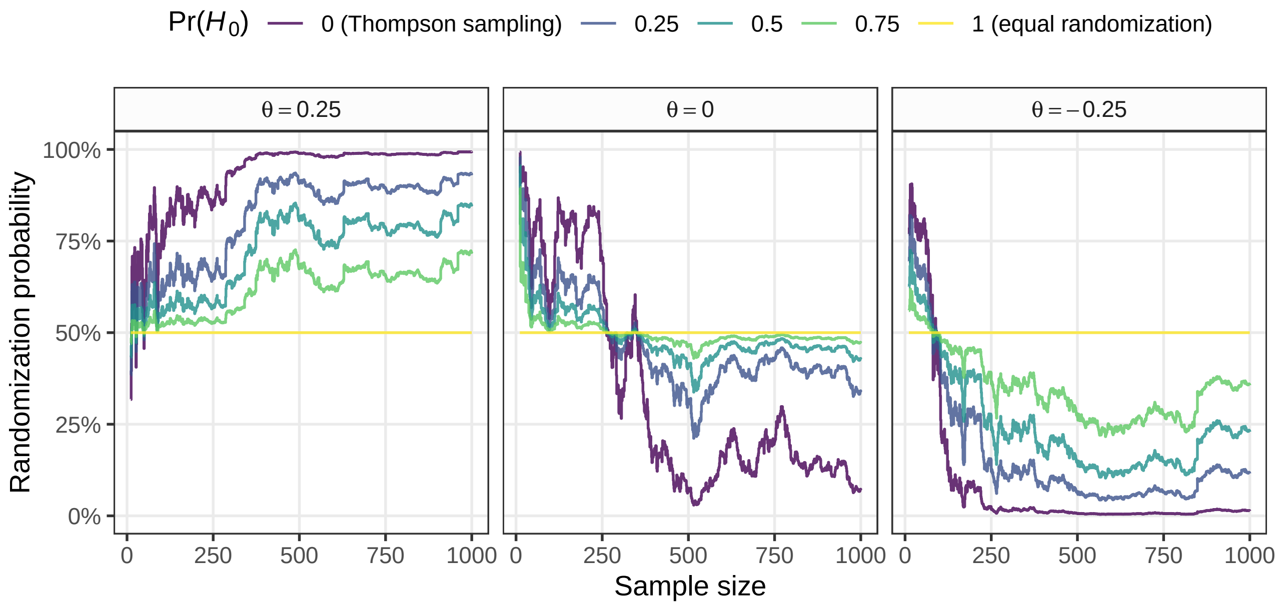

Figure 3: Evolution of Bayesian RAR probabilities for three simulated data sequences. In each step, one patient is allocated to the treatment or control group using Thompson sampling assuming a normal prior centered at zero with standard deviation $\tau = 1$. Patient outcomes are then simulated from a normal distribution with a true standard deviation of 1 and assuming a true mean difference θ between treatment and control group as indicated in the plot panels. Randomization probabilities under the spike-and-slab prior ($\Pr(H_0) > 0$) are computed from the same data sequence to enable comparability.

Figure 3 shows sequences of randomization probabilities computed from normal data simulated under different values of the true mean difference θ between treatment and control group as indicated in the plot panels. For each θ, one sequence was simulated under RAR using Thompson sampling ($\Pr(H_0) = 0$). RAR probabilities for other values of $\Pr(H_0)$ were computed from the same sequences to base probability comparison on the same data. Across all true mean differences θ, we can see that the probability to randomize to the treatment group based on $\Pr(H_0) = 1$ remains static at 50% (yellow line). In contrast, in the case of a beneficial treatment effect ($\theta > 0$; left plot), the randomization probabilities based on $\Pr(H_0) < 1$ tend towards 100% as more data accumulate, whereas in case of a harmful treatment effect ($\theta > 0$; right plot), they tend towards 0%. Moreover, a clear ordering is visible: Probabilities under $\Pr(H_0) = 0$, which corresponds to Thompson sampling, are the most extreme both in the "correct" and "wrong" direction. Probabilities based on $0 < \Pr(H_0) < 1$ show the same qualitative behavior but are less extreme. Setting a higher prior probability

$\Pr(H_0)$ thus reduces the variability of randomization probabilities but also makes convergence to 100% or 0% slower. Finally, when there is no treatment effect ($\theta = 0$; middle plot), the probabilities based on $0 < \Pr(H_0) < 1$ stay relatively close to 50%, whereas the Thompson sampling probabilities wander more erratically.

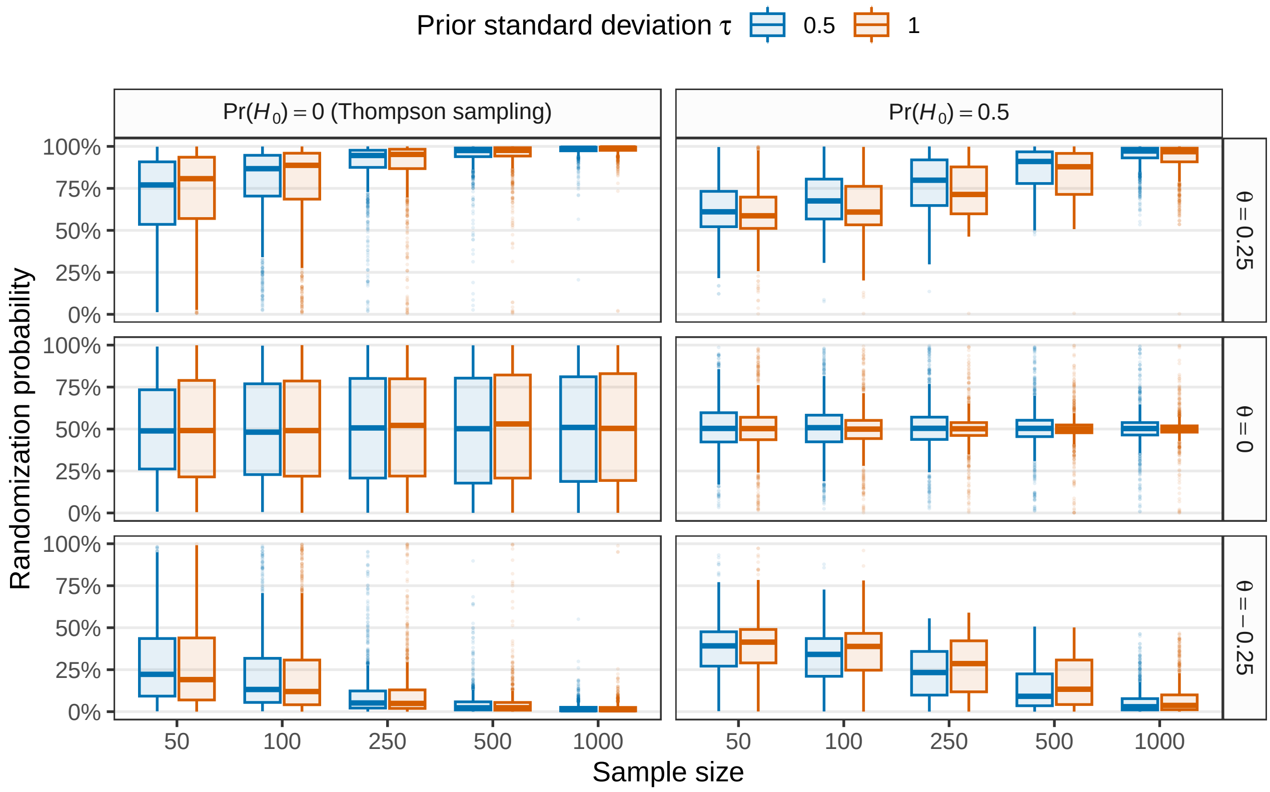

Figure 4: Distribution of randomization probabilities based on 1000 simulations of the scenarios from Figure 3.

Figure 4 shows the distribution of randomization probabilities obtained from 1000 simulations of the scenarios from Figure 3 and for two different prior standard deviations τ. In conditions where the true mean difference differs from zero ($\theta = \pm 0.25$; first and third row), we see that the randomization probabilities for both $\Pr(H_0) = 0$ (Thompson sampling) and $\Pr(H_0) = 0.5$ correctly converge to to 100% and 0%, respectively, although the convergence of $\Pr(H_0) = 0.5$ is slower. In contrast, when there is no true difference ($\theta = 0$; second row), only the randomization probabilities based on $\Pr(H_0) = 0.5$ converge to 50%, whereas the Thompson sampling randomization probabilities remain roughly uniformly distributed for all sample sizes. The qualitative behavior of the randomization probabilities is the same for both values of the prior standard deviation τ. However, convergence to 100% and 0% is slightly faster for the smaller prior standard deviation (blue), whereas convergence to 50% is faster for the larger prior standard deviation (orange).

3.2 More than two groups

Suppose now there are $K > 1$ treatment groups and consequently K effect estimates, each estimate quantifying the effect of the corresponding treatment relative to the control. A natural generalization is to stack the estimates into a vector $\hat{\boldsymbol{\theta}} = (\hat{\theta}_1, \dots, \hat{\theta}_K)^\top$ and assume an approximate K-variate normal distribution $\hat{\boldsymbol{\theta}} \mid \boldsymbol{\theta} \sim \mathrm{N}_K(\boldsymbol{\theta}, \boldsymbol{\Sigma})$, where $\boldsymbol{\theta} = (\theta_1, \dots, \theta_K)^\top$ is the vector of true effects and $\boldsymbol{\Sigma}$ is the (assumed to be known) covariance matrix of $\hat{\boldsymbol{\theta}}$. For example, $\hat{\boldsymbol{\theta}}$ could be a vector of estimated regression coefficients and $\boldsymbol{\Sigma}$ the corresponding covariance matrix.

The hypotheses from Section 2.2 then translate into

$$H_-\colon \max\{\theta_1, \dots, \theta_K\} < 0 \quad \text{vs.} \quad H_0\colon \theta_1 = \cdots = \theta_K = 0 \quad \text{vs.} \quad H_{+i}\colon \theta_i = \max\{\theta_1, \dots, \theta_K\} > 0$$

for $i = 1, \dots, K$. All hypotheses apart from H_0 are composite and require the specification of a prior distribution. In analogy to the one treatment case, we specify a K-variate normal prior $\boldsymbol{\theta} \sim \mathrm{N}_K(\boldsymbol{\mu}, \boldsymbol{\mathcal{T}})$ and truncate its support to the region of the corresponding hypothesis (e.g., for hypothesis H_{+i}, the space in $\mathbb{R}^K$ where the ith component is positive and larger than and all other components). As in the one group case, it seems a sensible default to center the prior on $\mathbf{0}$ to encode clinical equipoise. Similarly, the prior hypothesis probabilities $\Pr(H_{+i})$ for $i = 1, \dots, K$ may again be specified so that the $\mathrm{N}_K(\boldsymbol{\mu}, \boldsymbol{\mathcal{T}})$ distribution is recovered when the prior is averaged over the hypotheses.

Figure 5 illustrates such a spike-and-slab prior for the two treatment groups scenario ($K = 2$). Specifying a prior covariance matrix $\boldsymbol{\mathcal{T}}$ with uniform correlation of 0.5 ensures that the prior probabilities of all hypotheses but H_0 are equal. This can also be motivated by the fact that for normal outcomes with a shared control group and equal allocation, mean difference effect estimates are correlated by 0.5 due to the common control group for all

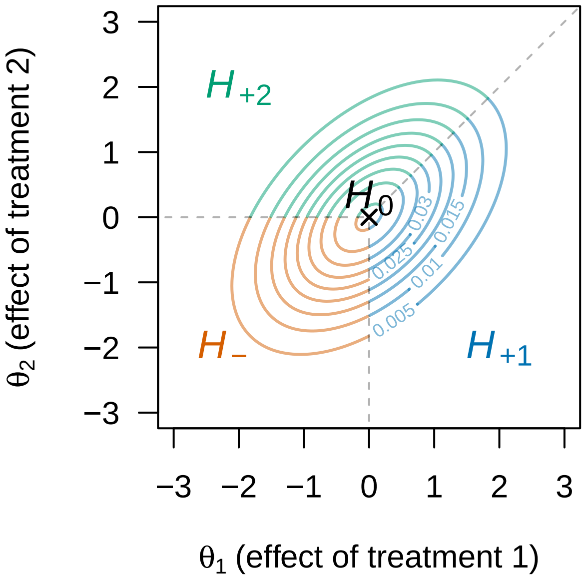

Figure 5: Illustration of a spike-and-slab prior for a two-dimensional effect $\boldsymbol{\theta} = (\theta_1, \theta_2)^\top$. A point mass prior at $(0,0)^\top$ is assumed under H_0. A normal prior $\theta \sim \mathrm{N}((0,0)^\top, \boldsymbol{\mathcal{T}})$ with $\boldsymbol{\mathcal{T}}_{ij} = 0.5$ for $i \neq j$ and $\boldsymbol{\mathcal{T}}_{ij} = 1$ for $i = j$, with support truncated to the space of the corresponding hypothesis is assumed under H_-, H_{+1}, and H_{+2}. The correlation ensures that all treatments receive equal prior probability $\Pr(H_-) = \Pr(H_{+1}) = \Pr(H_{+2}) = \{1 - \Pr(H_0)\}/3$.

treatments. Note that a zero-centered prior with covariance matrix $\boldsymbol{\mathcal{T}}$ with uniform correlation of 0.5 ensures equal treatment prior probabilities for any number of treatment groups K, and thus seems a sensible default in the absence of prior knowledege.

As in the two-group case (Section 3.1), the normal-normal conjugate framework allows us to derive marginal likelihoods in closed-form. The marginal likelihood under H_0 is

$$p(\hat{\boldsymbol{\theta}} \mid H_0) = \mathrm{N}_K(\hat{\boldsymbol{\theta}} \mid \mathbf{0}, \boldsymbol{\Sigma}),$$

while the marginal likelihood under H_- is

$$p(\hat{\boldsymbol{\theta}} \mid H_-) = \mathrm{N}_K(\hat{\boldsymbol{\theta}} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma} + \boldsymbol{\mathcal{T}}) \times \frac{\Phi_K(\mathbf{0} \mid \boldsymbol{\mu}_*, \boldsymbol{\mathcal{T}}_*)}{\Phi_K(\mathbf{0} \mid \boldsymbol{\mu}, \boldsymbol{\mathcal{T}})}$$

with $\mathrm{N}_K(\boldsymbol{x} \mid \boldsymbol{m}, \boldsymbol{V})$ and $\Phi_K(\boldsymbol{x} \mid \boldsymbol{m}, \boldsymbol{V})$ the density and cumulative distribution functions of the K-variate normal distribution with mean vector $\boldsymbol{m}$ and covariance matrix $\boldsymbol{V}$ evaluated at $\boldsymbol{x}$, and posterior mean $\boldsymbol{\mu}_* = (\boldsymbol{\Sigma}^{-1} + \boldsymbol{\mathcal{T}}^{-1})^{-1}(\boldsymbol{\Sigma}^{-1}\hat{\boldsymbol{\theta}} + \boldsymbol{\mathcal{T}}^{-1}\boldsymbol{\mu})$ and covariance $\boldsymbol{\mathcal{T}}_* = (\boldsymbol{\Sigma}^{-1} +$

$\boldsymbol{\mathcal{T}}^{-1})^{-1}$. Finally, the marginal likelihood under H_{+i} is given by

$$p(\hat{\boldsymbol{\theta}} \mid H_{+i}) = \mathrm{N}_K(\hat{\boldsymbol{\theta}} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma} + \boldsymbol{\mathcal{T}}) \times \frac{\Phi_K(\mathbf{0} \mid \boldsymbol{A}_{i,K}\boldsymbol{\mu}_*, \boldsymbol{A}_{i,K}\boldsymbol{\mathcal{T}}_*\boldsymbol{A}_{i,K}^\top)}{\Phi_K(\mathbf{0} \mid \boldsymbol{A}_{i,K}\boldsymbol{\mu}, \boldsymbol{A}_{i,K}\boldsymbol{\mathcal{T}}\boldsymbol{A}_{i,K}^\top)}$$

where $\boldsymbol{A}_{i,K}$ is a $K \times K$ contrast matrix that maps $\boldsymbol{\theta}$ to the space where the negative orthant corresponds to the space of hypothesis H_{+i}. For example, for $i = 2$ and $K = 3$, the matrix is

$$\boldsymbol{A}_{2,3} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & -1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

with the first row encoding the constraint of θ_2 being positive, and the second and third rows encoding the constraints of θ_2 being larger than θ_1 and θ_3, respectively. As expected, for $K = 1$, these marginal likelihoods reduce to (7). Moreover, since they are all available in closed-form, Bayes factors, posterior probabilities, and randomization probabilities are also available in closed-form. Crucially, only the exact evaluation of multivariate normal densities and cumulative distribution functions is required, and both are efficiently implemented in statistical software (e.g., in the `mvtnorm` R package, Genz and Bretz, 2009). This thus leads to an efficient Bayesian RAR method that allows experimenters to compute randomization probabilities in complex settings, for instance, multiple regression, where a full Bayesian analysis may involve various complexities, such as priors for nuisance parameters and Markov chain Monte Carlo methods for the computation of posteriors.

[TODO: this paragraph and figure could be removed for space?] Figure 6 shows sequences of randomization probabilities computed from simulated normal data with $K = 3$ treatment groups. We see again that assigning a prior probability $\Pr(H_0) = 1$ leads to static equal randomization at $\pi_i = 1/(K+1) = 25\%$, while assigning $\Pr(H_0) = 0$ (Thompson sampling) leads to the most variable randomization probabilities. Since data are simulated assuming that treatment 1 is the most effective, randomization probabilities based on $\Pr(H_0) < 1$ converge towards $\pi_1 = 100\%$ and toward 0% for the remaining treatments.

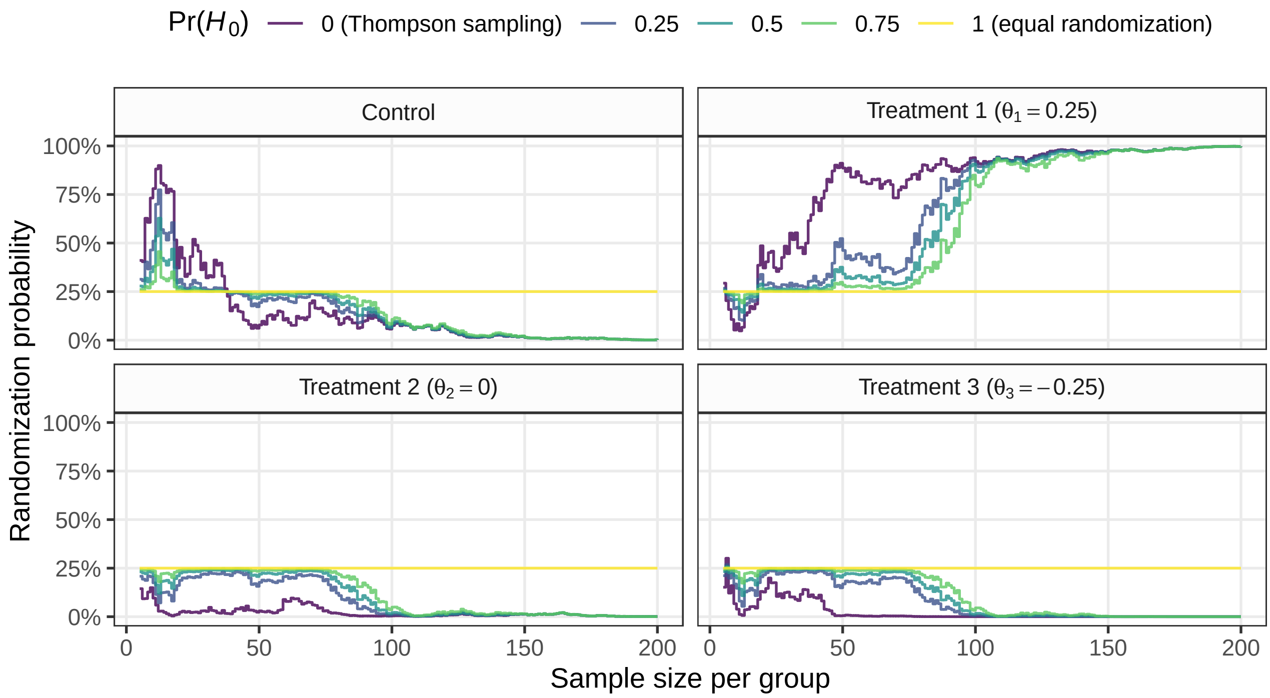

Figure 6: Evolution of Bayesian RAR probabilities for 3 treatments and a control group. In each step, one observation from each group is simulated from a normal distribution with a standard deviation of 1 and corresponding mean differences as indicated in the panel titles (treatment 1 is the most effective treatment), and a mean difference effect estimate $\hat{\theta}$ and its covariance matrix $\mathbf{\Sigma}$ computed with linear regression. Randomization probabilities are then computed using a normal spike-and-slab prior for centered at the origin and with covariance matrix $\mathcal{T}$ with $\mathcal{T}_{ij} = 0.5$ for $i \neq j$ and $\mathcal{T}_{ij} = 1$ for $i = j$, and for different prior probabilities $\Pr(H_0)$.

4 Null hypothesis Bayesian RAR for binary outcomes

We now consider the setting with binary outcomes, as such outcomes frequently occur in applications of Bayesian RAR (e.g., in clinical trials or A/B testing settings). Suppose that we observe data of the form $y = \{y_C, y_1, \dots, y_K, n_C, n_1, \dots, n_K\}$ where y_i denotes the number of successes out of n_i trials in group $i \in \{C, 1, \dots, K\}$ coming from a control group (index C) and K treatment groups. All success counts are assumed to be binomially distributed with probabilities $\theta_C, \theta_1, \dots, \theta_K$, respectively, and higher values are assumed to indicate a higher benefit (e.g., a higher probability of disease recovery). The hypotheses from Section 2.2 then translate into

$$H_-\colon \theta_C = \max\{\theta_C, \theta_1, \dots, \theta_K\} \quad \text{vs.} \quad H_0\colon \theta_C = \theta_1 = \cdots = \theta_K \quad \text{vs.} \quad H_{+i}\colon \theta_i = \max\{\theta_C, \theta_1, \dots, \theta_K\}$$

for $i = 1, \dots, K$. In the approximate normal framework from Section 3, these hypotheses could be translated into hypotheses related to log odds ratios $\psi_i = \log[\{\theta_i(1-\theta_C)\}/\{(1-\theta_i)\theta_C\}]$, which can be estimated with logistic regression or other methods (we will provide a comparison with this approach below). However, such normal approximations can be inaccurate for small sample sizes and/or extreme probabilities close to zero/one. It is therefore preferable to compute randomization probabilities via the exact binomial distribution.

To compute posterior and randomization probabilities, it is necessary to compute the marginal likelihood of the observed data y under the different hypotheses. The null hypothesis H_0 requires specification of a prior for the common probability θ_C. Assuming a beta prior $\theta_C \mid H_0 \sim \text{Beta}(a_0, b_0)$, the marginal likelihood of the observed data under H_0 is

$$\Pr(y \mid H_0) = \prod_{j \in \{C,1,\dots,K\}} \binom{n_j}{y_j} \times \frac{\mathrm{B}(a_0 + \sum_{i \in \{C,1,\dots,K\}} y_i, b_0 + \sum_{i \in \{C,1,\dots,K\}} n_i - \sum_{i \in \{C,1,\dots,K\}} y_i)}{\mathrm{B}(a_0, b_0)}$$

with $\mathrm{B}(a,b) = \int_0^1 t^{a-1}(1-t)^{b-1}\mathrm{d}t$ the beta function. Under the remaining hypotheses, it is natural to assume independent beta priors $\theta_i \sim \text{Beta}(a_i, b_i)$ for $i \in \{C, 1, \dots, K\}$, and truncate their support to the space of the corresponding hypothesis. This leads to the marginal likelihoods under hypothesis $H_i \in \{H_C, H_{+1}, \dots, H_{+K}\}$

$$\begin{aligned} \Pr(y \mid H_i) = &\prod_{j \in \{C,1,\dots,K\}} \binom{n_j}{y_j} \times \frac{\mathrm{B}(a_j + y_j, b_j + n_j - y_j)}{\mathrm{B}(a_j, b_j)} \\ &\times \frac{Q_i(a_C + y_C, a_1 + y_1, \dots, a_K + y_K, b_C + n_C - y_C, b_1 + n_1 - y_1, \dots, b_K + n_K - y_K)}{Q_i(a_C, a_1, \dots, a_K, b_C, b_1, \dots, b_K)} \end{aligned} \tag{9}$$

with

$$Q_i(a_C, a_1, \dots, a_K, b_C, b_1, \dots, b_K) = \int_0^1 \frac{\theta_i^{a_i - 1}(1-\theta_i)^{b_i - 1}}{\mathrm{B}(a_i, b_i)} \times \prod_{j \in \{C,1,\dots,K\} \setminus \{i\}} I_{\theta_i}(a_j, b_j)\, \mathrm{d}\theta_i,$$

where $I_x(a,b) = \{\int_0^x t^{a-1}(1-t)^{b-1}\,\mathrm{d}t\}/\mathrm{B}(a,b)$ is the regularized incomplete beta function, also known as the cumulative distribution function of the beta distribution. These can be efficiently computed with numerical integration and standard implementations of the regularized incomplete beta function (e.g., `stats::pbeta` in R). For priors with integer hyperparameters and up to four treatment groups, there even exist closed-form solutions in terms of beta functions (Howard, 1998; Miller, 2015). For example, for one treatment group ($K = 1$), we have that $Q_1(a_C, a_1, b_C, b_1) = \sum_{j=0}^{a_1-1} \mathrm{B}(a_C + j, b_C + b_1)/\{(b_1 + j)\mathrm{B}(j + 1, b_1)\mathrm{B}(a_C, b_C)\}$.

For a specified prior probability of the null hypotheses $\Pr(H_0)$, we may again distribute the remaining prior probability among the other hypotheses by $\Pr(H_{+i}) = \{1 - \Pr(H_0)\} \times Q_i(a_C, a_1, \ldots, a_K, b_C, b_1, \ldots, b_K)$ to ensure that the averaged prior is again the beta prior that was truncated in the first place. Specifying uniform priors ($a_j = b_j = 1$ for $j = 1, \ldots, K$) seems an intuitive default that ensures that each treatment receives equal prior probabilitiy. Plugging these marginal likelihoods and prior probabilities into equation (1) produces posterior probabilities, which in turn can be used for obtaining randomization probabilities. Similarly, ratios of marginal likelihoods can be taken to obtain Bayes factors for monitoring accumulating evidence.

5 Reanalyzing the ECMO trial

The ECMO trial (Bartlett et al., 1985) investigated the efficacy of ECMO (extracorporeal membrane oxygenation) treatment in the critical care of newborns. It was the first prominent clinical trial to use a "randomized play-the-winner" (RPW) RAR design (Wei and Durham, 1978). While some earlier non-randomized studies had shown a substantial treatment effect, a randomized study was required to confirm this. However, this presented an ethical dilemma, as the investigators were convinced that the risk of death would be much higher in the control group than in the ECMO group. To mitigate this, the RPW design was chosen.

The outcome of the trial was extreme: The first newborn was randomized to receive ECMO treatment and survived. The second newborn was randomized to receive the control treatment and died. The ten subsequently enrolled newborns were all randomized to ECMO, and all survived (Bartlett et al., 1985). The trial was then stopped for efficacy and its results published. However, the unusual outcome sparked intense debates about whether the trial was even a proper randomized clinical trial. Several subsequent trials were con-

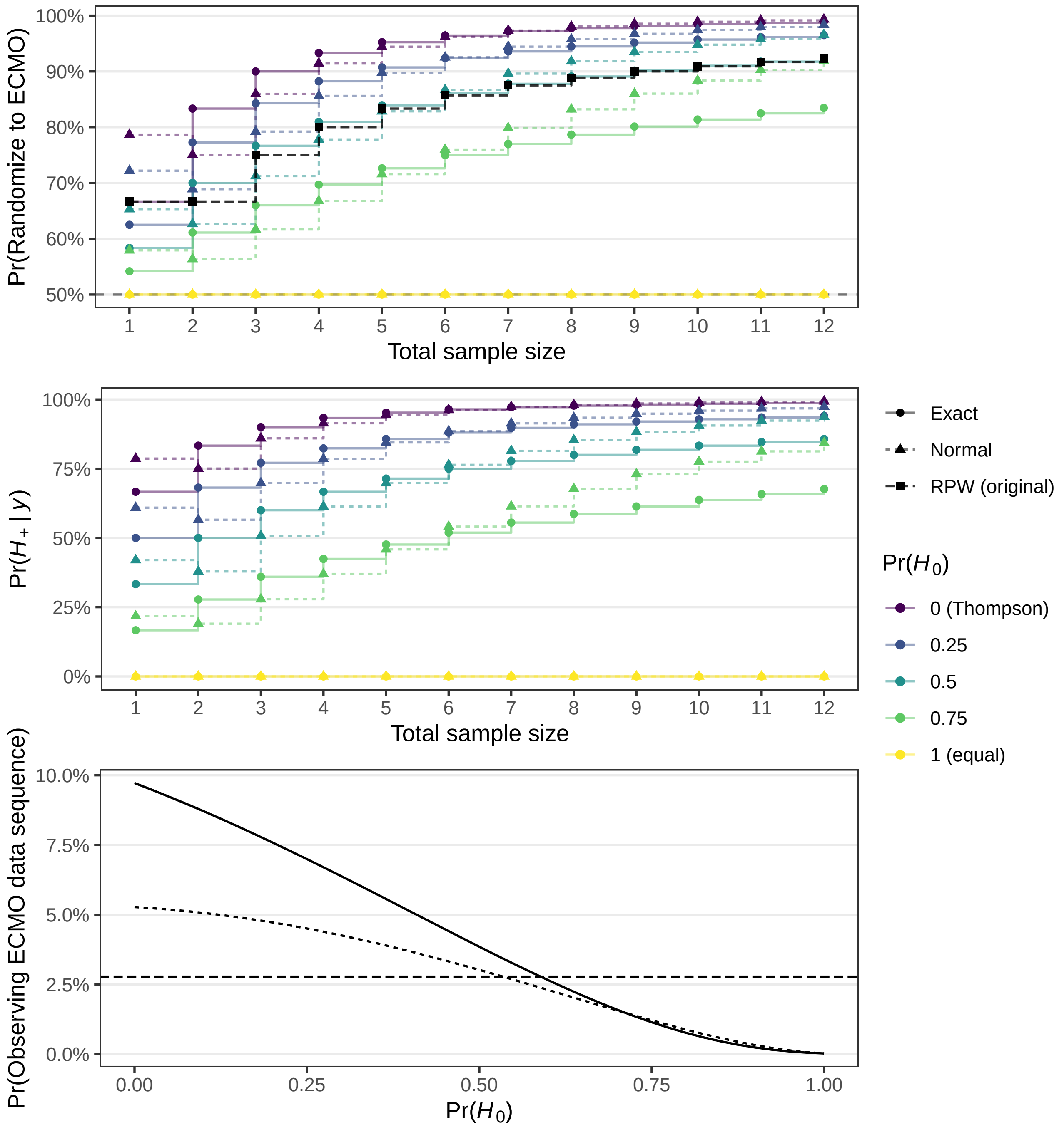

Figure 7: Evolution of Bayesian RAR randomization probabilities (top plot) and posterior probability of a beneficial ECMO treatment effect (middle plot) for data from the ECMO trial. The bottom plot shows the probability of observing the ECMO data sequence as a function of $\Pr(H_0)$, the long-dashed horizontal line shows the corresponding probability under the randomized play the winner (RPW) rule used in the ECMO trial.

ducted, all re-establishing the effectiveness of ECMO, which is now a standard treatment (Bartlett, 2024). In the following, we will reanalyze the ECMO data.

The top plot in Figure 7 shows Bayesian RAR probabilities computed from the ECMO data sequence. The normal approximation and the exact binomial methods were used, as well as the RPW method that was originally used in the ECMO trial. A standard normal prior was assigned to the log odds ratio for the normal method, while uniform priors were assigned to the probabilities for the binomial method. For the normal method, log odds ratio estimates were computed with a Yates' correction (i.e., adding a half to each cell) to avoid issues with zero cells. We can see that this correction induces a slight anomaly at the start of the study as the probabilities from the normal method slightly decrease after the first observation despite the first patient surviving the ECMO treatment. This does not happen for the exact method whose randomization probability increase after every additional patient, which only happens for the normal approximation after the second patient. Furthermore, the randomization probabilities differ notably between the normal and exact methods, even at the end of the study. This presumably happens because the influence of the differing priors remains relatively high after observing only 12 patients.

Comparing the randomization probabilities for different prior probabilities of the null hypothesis for each method, we can see that $\Pr(H_0) = 0$ (Thompson sampling) shows the most extreme randomization probabilities that rapidly increase to 100%, whereas $\Pr(H_0) = 1$ (equal randomization) leaves the probabilities completely static at 50%. In between, the randomization probabilities gradually shift from 50% to the Thompson sampling probabilities. The RPW probabilities (black squares) are relatively close to the probabilities from the exact method with $\Pr(H_0) = 0.5$ at the beginning of the study and become closer to the normal approximate method with $\Pr(H_0) = 0.75$ at later time points.

The middle plot in Figure 7 shows the corresponding posterior probability of the ECMO treatment being effective. Depending on the prior probability $\Pr(H_0)$, we can see that the posterior probability at the end of the trial may be very high (e.g., $\Pr(H_+ \mid y) = 0.99$ for $\Pr(H_0) = 0$), or only moderately favoring ECMO (e.g., $\Pr(H_+ \mid y) = 0.86$ for $\Pr(H_0) = 0.5$). From a Bayesian perspective, stopping the trial seems thus only a sensible decision if the prior probability of equal effects was low. Conversely, more evidence would have been needed if the prior probability had been higher, for example, if the prior probability were $\Pr(H_0) = 0.5$, representing a priori equipoise.

Finally, the bottom plot in Figure 7 shows the probability of observing the ECMO data sequence under Bayesian RAR for different prior probabilities $\Pr(H_0)$. As expected, the probability is highest for Thompson sampling ($\Pr(H_0) = 0$) and decreases with increasing $\Pr(H_0)$. Interestingly, the data probability under the RPW method used in the ECMO trial is rather low (2.8%; horizontal long-dashed line), and observing the data under Bayesian RAR with $\Pr(H_0)$ about below 0.6 would be more likely.

6 Simulation study

We conducted a simulation study to evaluate the performance of null hypothesis Bayesian RAR for different values of the prior probability of the null hypothesis $\Pr(H_0)$, and compare it to Thompson sampling (potentially modified with burn-in periods, probability capping, power transformations), equal randomization, and two methods from the multi-armed bandit literature – the Gittins index (Gittins et al., 2011; Villar et al., 2015a) and Bayes upper confidence bound (UCB) (Kaufmann et al., 2012). Considered patient performance measures were the rate of successes, the rate of extreme randomization probabilities, and sample size imbalance. Additionally, bias and coverage were considered to evaluate the performance of response rate difference point estimates and confidence intervals under RAR, while the type I error rate and power of the corresponding tests were used to quantify hypothesis testing performance under RAR. A binomial data-generating mechanism was used which was partially based on the simulation studies from Robertson et al. (2023), Thall and Wathen (2007), and Wathen and Thall (2017). The response probability in the control group was fixed to $\theta_C = 0.25$ and varied in the first treatment group $\theta_1 \in \{0.25, 0.35, 0.45\}$ to assess the behavior of the method for different effect sizes $\mathrm{RD}_1 = \theta_1 - \theta_C$. In conditions with two or three treatment groups, the corresponding probabilities were set to $\theta_2 = \theta_3 = 0.3$. More detailed descriptions of the design and results of the simulation study following the structured ADEMP approach (Morris et al., 2019; Siepe et al., 2024) are provided in Appendix B. A supplemental website provides an interactively explorable results dashboard (`https://samch93.github.io/brar/`).

Across most simulations, we observed a trade-off between patient benefit and inferential operating characteristics. Methods that improved patient benefit through more aggressive allocation (e.g., Gittins index, Bayes UCB, and Thompson sampling) often had worse bias, coverage, type I error rate, and power, whereas more conservative methods tended to preserve inferential validity at the cost of reduced patient benefit. Figure 8 and Figure 9 illustrate this trade-off for the mean success rate and confidence interval coverage performance measures, respectively. This trade-off is well described in the RAR literature (Hu and Rosenberger, 2003; Zhang and Rosenberger, 2005; Williamson and Villar, 2019; Robertson et al., 2023). However, there were also settings in which there was a small "window" in which patient benefit could vary while inferential performance remained essentially unchanged. For example, with a burn-in of 50, two treatment groups ($K = 2$), and a large effect size ($\mathrm{RD}_1 = 0.2$), confidence interval coverage was roughly constant across values of $\Pr(H_0)$, whereas the mean success rate changed noticeably. Furthermore, increasing $\Pr(H_0)$ reduced sample size imbalance in direction of the inferior treatment compared to Thompson sampling (Figure 12 in Appendix B). Yet, negative imbalance was also strongly influenced by the effect size RD_1. In conditions with large effect size ($\mathrm{RD}_1 = 0.2$), the proportion of negative imbalance was often lower for RAR methods (including Thompson sampling) than equal randomization, thus replicating the results from Robertson et al. (2023). These findings highlight the importance of evaluating operating characteristics during trial design, since appropriately specified RAR procedures may enable improved patient benefit without substantial loss of inferential performance.

The main result regarding the newly proposed method was that, under most conditions, null hypothesis Bayesian RAR with a high prior probability of the null hypothesis $\Pr(H_0) = 0.75$ showed similar operating characteristics to Thompson sampling with capped randomization probabilities at 10% and 90% and a power transformation with $c = i/(2n)$, where i is the current sample size and n is the maximum sample size. Both Bayesian RAR with a prior probability of $\Pr(H_0) = 0.75$ and modified Thompson sampling could mitigate some issues with ordinary Thompson sampling. For instance, they exhibited less negative sample size imbalance, less biased parameter estimates, improved coverage, and reduced

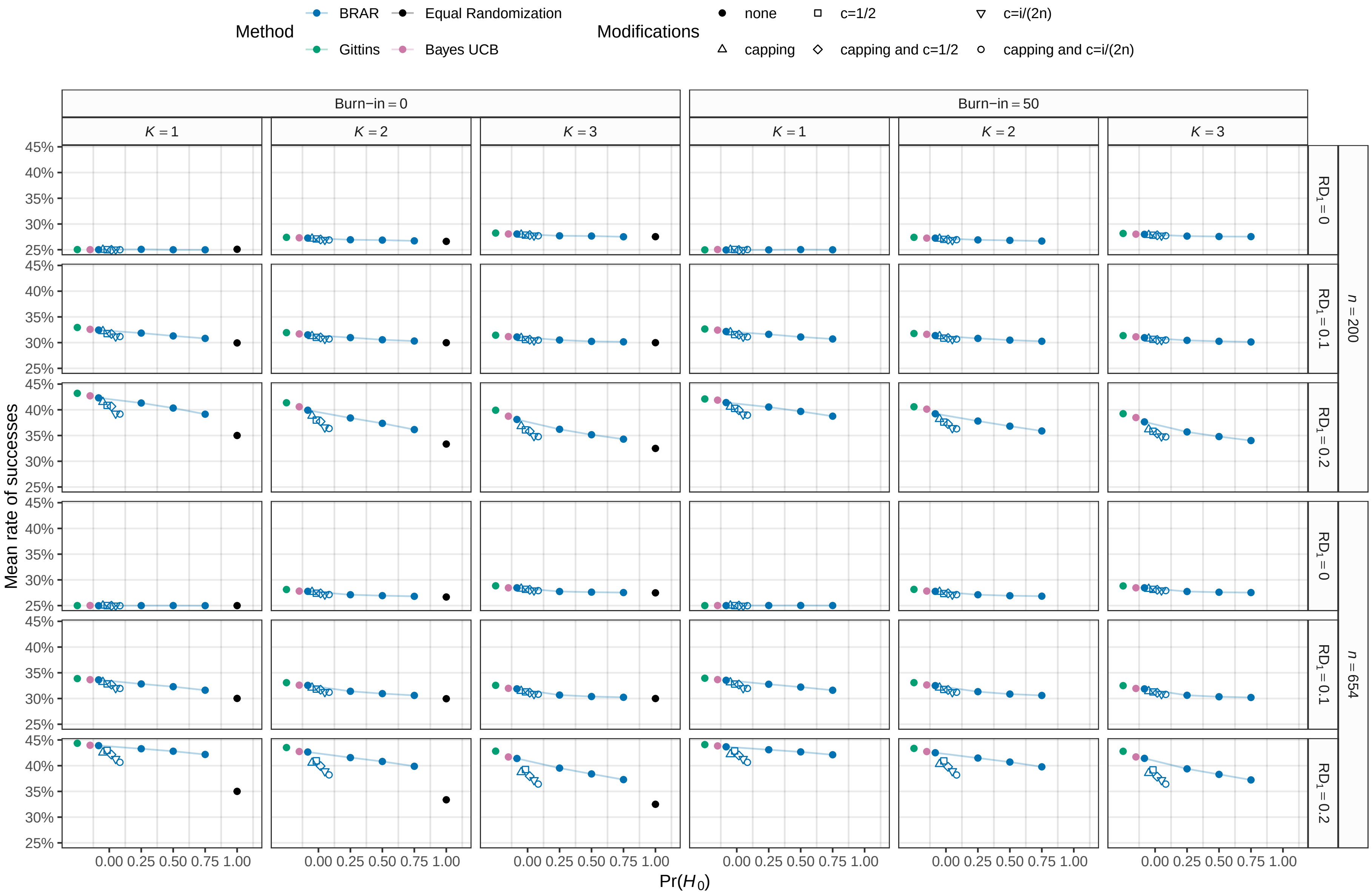

Figure 8: Mean rate of successes (i.e., the number of successes in a study divided by its sample size averaged over all 10′000 simulation repetitions). The maximum MCSE is 0.051%.

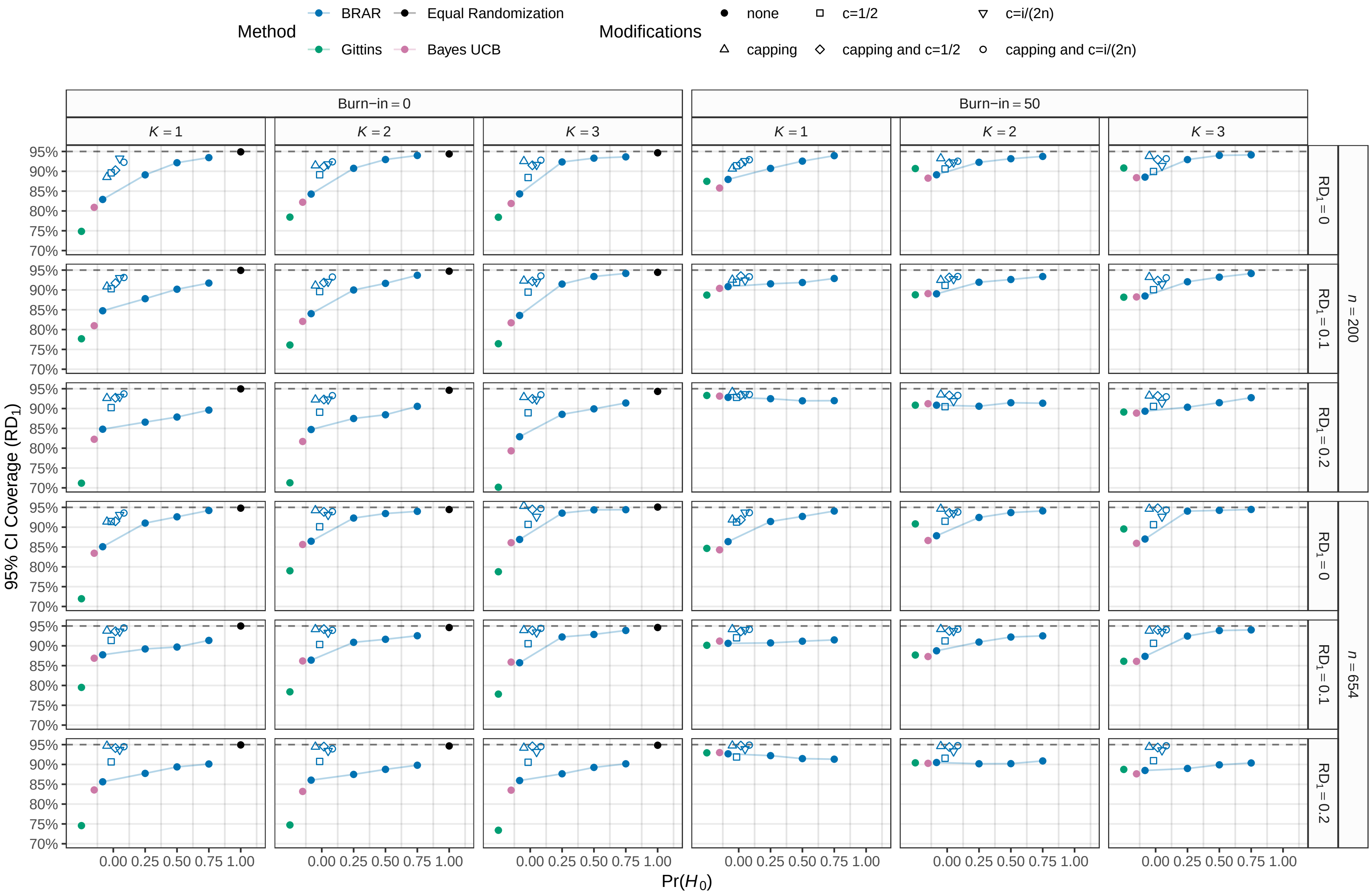

Figure 9: Empirical coverage of the 95% Wald confidence interval for the rate difference RD_1 based on 10′000 simulation repetitions. The maximum MCSE is 0.46%.

type I error rates. However, this improvement came at the cost of worse patient benefit performance compared to unmodified Thompson sampling. For instance, the mean success rate was lower, though still considerably better than equal randomization in most cases. Setting a lower but positive prior probability than $\Pr(H_0) = 0.75$ produced operating characteristics comparable to those of less extreme Thompson sampling modifications, such as a power transformation with $c = 1/2$. Finally, the simulation study was performed for both the exact binomial and the approximate normal version of the method (shown only in Appendix B). Despite using different likelihoods and priors, the results were similar, indicating some robustness of our conclusions with respect to priors and models.

While the simulation study demonstrated that null hypothesis Bayesian RAR has comparable statistical properties to Thompson sampling with common ad hoc modifications in realistic trial sizes, it is important to emphasize that these methods are expected to differ asymptotically. The allocated proportion of participants under null hypothesis Bayesian RAR can converge to 100% or 0% if there is an actual treatment effect, which is not the case for capped Thompson sampling which enforces fixed lower and upper bounds on allocation probabilities. Thus, the observed similarity should be interpreted as a finite-sample phenomenon, where the introduction of the null hypothesis H_0 induces more conservative early allocation.

7 Discussion

In this paper we have proposed a modification of standard Bayesian RAR (Thompson sampling) via recasting of the problem in a Bayesian hypothesis testing framework and the introduction of a null hypothesis of equal effectiveness. This method is useful in the RAR setting, as it can interpolate between equal randomization and Thompson sampling by changing the prior probability of the null hypothesis $\Pr(H_0)$. This allows experimenters to balance patient benefit with inferential operating characteristics, such as power, type I error rate, bias, and coverage. For large values of $\Pr(H_0)$, we observed behaviors and operating characteristics similar to those obtained with ad hoc modifications of Thompson sampling, such as capping, burn-ins, and power transformations. Our method is implemented in the free and open source R package `brar` for binomial outcomes and for data summarized by approximately normal effect estimates. The latter makes the method applicable to many settings, for example, settings where treatment effects are estimated with regression analyses.

One advantage of our framework is that the randomization probabilities coherently correspond with the available statistical evidence (in the form of Bayes factors) and beliefs (in the form of posterior probabilities). In principle, both could also be used as decision-making tools instead of relying on frequentist test criteria. For instance, if the posterior probability of a treatment's superiority is greater than 0.99, say, a study could be stopped. This also makes sense from the perspective that it is unnatural to randomize participants with extreme randomization probabilities that are associated with such high posterior probabilities.

Although we conducted a simulation study to understand the method's basic behavior, more realistic and comprehensive evaluations are needed to understand its applicability in real-world conditions. For example, the method needs to be evaluated in combination with futility stopping (e.g., dropping of treatment arms which are shown to be ineffective at an interim analysis) and under time trends (e.g., treatment effects attenuating over time because the standard of care in the control group improves). Another issue to consider is how to select the prior probability of the null hypothesis. In our simulation study, we found that setting a value of $\Pr(H_0) = 0.75$ mitigated many of the issues with Thompson sampling. However, other choices could be considered, such as setting a higher value. Similar considerations apply to the prior distributions of the parameters, as we only limitedly assess the effect of varying these on the operating characteristics. For instance, rather than setting the correlation of the multivariate normal prior to achieve equal randomization probabilities, it could be beneficial to set it so that the prior probability of the control being superior is always $\Pr(H_-) = 0.5$ and the remaining probability is distributed equally among the treatments. Future work may therefore investigate whether a more efficient RAR procedure can be obtained by specifying a certain prior distribution. Beyond simulation, also further theoretical analysis is needed to understand the asymptotic behavior of the method. Most importantly, it remains to be rigorously shown that the randomization probability based on

$\Pr(H_0) > 0$ converges to 50% as the sample size increases when the data are generated under H_0, as we found empirically. In many clinical trial settings, RAR methods are not directly applicable because outcomes such as death may only be observed after long follow-up periods, by which time recruitment and randomized allocation will already have finished. An alternative could be to perform Bayesian RAR with an informative surrogate outcome that is sooner observed than the primary outcome (Gao et al., 2024). Finally, in the multi-arm case it may be desirable to ensure concurrent allocation to the control group, for example, the method from Trippa et al. (2012) ensures that the number of patients assigned to the control group approximately equals the the experimental arm with the highest sample size. It would be interesting to explore whether the method can be modified to take concurrent allocation into account.

Acknowledgments and conflict of interest

We thank František Bartoš and Małgorzata Roos for valuable comments on drafts of the manuscript. We thank the associate editor and two anonymous reviewers for excellent suggestions that considerably improved the paper. The acknowledgment of these individuals does not imply their endorsement of the paper. We declare no conflict of interest.

Software and data

Code and data to reproduce our analyses are openly available at `https://github.com/SamCH93/brar`. A snapshot of the repository at the time of writing is available at `https://doi.org/10.5281/zenodo.17248628`. We used the statistical programming language R version 4.5.2 (2025-10-31) for analyses (R Core Team, 2024) along with the `ggplot2` (Wickham, 2016), `dplyr` (Wickham et al., 2023), `SimDesign` (Chalmers and Adkins, 2020), `mvtnorm` (Genz and Bretz, 2009), `ggh4x` (van den Brand, 2024), `ggpubr` (Kassambara, 2023), `RARtrials` (Xu et al., 2025), and `knitr` (Xie, 2015) packages.

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Appendix A The R package brar

Our R package can be installed by running `remotes::install_github(repo = "SamCH93/brar", subdir = "package")` in an R session (requires the `remotes` package, which is available on CRAN). The main functions of the package are `brar_normal` and `brar_binomial`, which implement the approximate normal method from Section 3 and the exact binomial method from Section 4. The following code chunk illustrates how the latter function can be used.

```
library(brar) # load package

## observed successes and trials in control and 3 treatment groups
y <- c(10, 9, 14, 13)
n <- c(20, 20, 22, 21)

## conduct exact point null Bayesian RAR
brar_binomial(y = y, n = n,
              ## uniform prior for common probability under H0
              a0 = 1, b0 = 1,
              ## uniform priors for all probabilities
              a = c(1, 1, 1, 1), b = c(1, 1, 1, 1),
              ## prior probability of the null hypothesis
              pH0 = 0.5)

## DATA
##              Events Trials Proportion
## Control          10     20      0.500
## Treatment 1       9     20      0.450
## Treatment 2      14     22      0.636
## Treatment 3      13     21      0.619
##
## PRIOR PROBABILITIES
##    H-    H0   H+1   H+2   H+3
## 0.125 0.500 0.125 0.125 0.125
##
## BAYES FACTORS (BF_ij)
##         H-     H0   H+1    H+2   H+3
## H-   1.000 0.0341  2.16 0.1837 0.223
## H0  29.335 1.0000 63.45 5.3891 6.533
## H+1  0.462 0.0158  1.00 0.0849 0.103
## H+2  5.443 0.1856 11.77 1.0000 1.212
## H+3  4.490 0.1531  9.71 0.8249 1.000
##
## POSTERIOR PROBABILITIES
##      H-      H0     H+1     H+2     H+3
## 0.00777 0.91148 0.00359 0.04228 0.03488
##
## RANDOMIZATION PROBABILITIES
```

```
##     Control Treatment 1 Treatment 2 Treatment 3
##       0.236       0.231       0.270       0.263
```

Appendix B Simulation study

We now describe the design and results of our simulation study following the structured ADEMP approach (Morris et al., 2019; Siepe et al., 2024). Our simulation study was not preregistered as it constitutes early-phase methodological research where the properties of a new method are explored without the intention to give wide recommendations for practitioners (Heinze et al., 2023). A website with additional details and results is provided at `https://samch93.github.io/brar/`.

B.1 Aims

The aim of the simulation study is to evaluate the design characteristics of the newly proposed null hypothesis Bayesian RAR approach, and compare it to existing methods.

B.2 Data-generating mechanism

The data-generating mechanism was inspired by the simulation studies from Robertson et al. (2023), Thall and Wathen (2007), and Wathen and Thall (2017). In each repetition, a data set with n binary outcomes is simulated through RAR: A patient i is randomly allocated to the control group or one of the K treatment groups based on randomization probabilities computed from the $1, \dots, i-1$ preceding outcomes. Depending on the allocation, an outcome is either simulated from a Bernoulli distribution with probability θ_C in the control group, θ_1 in the first treatment group, or θ_2 for the remaining treatment groups (in case $K > 1$).

Parameters were chosen similar to the simulation study from Robertson et al. (2023). We vary the sample size $n \in \{200, 654\}$ to represent low and high powered studies, the number of treatment groups $K \in \{1, 2, 3\}$, and the probability in the first treatment group $\theta_1 \in \{0.25, 0.35, 0.45\}$. The probability in the control group and the remaining groups is always fixed at $\theta_C = 0.25$ and $\theta_2 = \theta_3 = 0.3$, respectively. All these parameters are varied fully-factorially, leading to $2 \times 3 \times 3 = 18$ parameter conditions.

Since treatment allocation determines from which true probability an outcome is simulated, data generation is directly influenced by the RAR methods described below. These come with additional parameters that are, however, considered as method tuning parameters rather than true underlying parameters.

B.3 Estimands and other targets

The primary interest of this simulation study lies in assessing the patient benefit characteristics of different RAR methods. Additionally, the estimand of interest is the response rate difference $\mathrm{RD}_1 = \theta_1 - \theta_C$ and the target of interest is the null hypothesis of $\mathrm{RD}_1 = 0$.

B.4 Methods

We consider the null hypothesis Bayesian RAR method described in the main text. The prior probability of H_0 is a tuning parameter and controls the variability of the randomization probabilities. Setting $\Pr(H_0) = 1$ produces equal randomization, whereas $\Pr(H_0) = 0$ produces Thompson sampling. We consider values of $\Pr(H_0) \in \{0, 0.25, 0.5, 0.75, 1\}$, as well as the normal approximation and exact binomial version of RAR. For approximate normal RAR, a normal prior with mean 0, variance 1, and in case of $K > 1$ a correlation of 0.5, is considered. Independent uniform priors are assigned for binomial RAR. Log odds ratios along with their covariance are estimated with logistic regression and then used as inputs for the normal RAR method, while the exact method uses success counts and sample sizes only. In case a method fails to converge, equal randomization is applied as a back-up strategy, as this mimics what an experimenter might do in practice when a RAR method fails to converge (Pawel et al., 2025).

We also consider three modifications of these methods: In some conditions, a “burn-in” phase is carried out during which the first 50 patients are always randomized with equal probability $1/(K+1)$ to each group. For Thompson sampling ($\Pr(H_0) = 0$), we additionally consider conditions with power transformations of randomization probabilities, i.e., if π_k is the randomization probability of group k, we take $\pi_k^* = \pi_k^c / \sum_{j \in \{C,1,\ldots,K\}} \pi_j^c$. We consider $c = 1/2$ and $c = i/(2n)$ with i the current and n the maximal sample size, which are two popular choices of the tuning parameter c (Wathen and Thall, 2017). Additionally, in some conditions "capping" is applied to Thompson sampling. That is, randomization probabilities outside the $[10\%, 90\%]$ interval are set to either 10% or 90%. After capping has been performed, randomization probabilities are re-normalized to sum to one (Wathen and Thall, 2017; Lee and Lee, 2021). This re-normalization is only performed for randomization probabilities greater than 10%, as these would otherwise be reduced again to probabilities less than 10%. In case, a re-normalized probability becomes less than 10%, it is also capped at 10% and a second re-normalization performed. For equal randomization ($\Pr(H_0) = 1$), no burn-in, capping, or power transformation conditions are simulated as these manipulations have no effect.

Finally, upon suggestions from the associate editor, we also include two methods from the multi-armed bandit literature in the simulation study to compare null hypothesis Bayesian RAR to more aggressive allocation methods – the Gittins index (Gittins et al., 2011; Villar et al., 2015a) and the Bayesian upper confidence bound (Bayes UCB) methods (Kaufmann et al., 2012). For the Gittins index, we use the implementation from the `RARtrials` R package (Xu et al., 2025) with a discount factor of $d = 0.995$. For Bayes UCB we use uniform priors for each probability along with the tuning parameter $c = 0$, as recommended in Kaufmann et al. (2012) based on their simulation results.

B.5 Performance measures

Patient benefit was quantified with:

- The mean rate of successes per study

$$\overline{\mathrm{RS}} = \frac{1}{n_{\mathrm{sim}}} \sum_{i=1}^{n_{\mathrm{sim}}} \sum_{j=1}^{n} \frac{y_{ij}}{n}$$

  where y_{ij} denotes the 0/1 success indicator of patient j in simulation i, n is the sample size, and n_{sim} is the number of simulation repetitions.

- The mean rate of extreme randomization probabilities (less than 10% or greater than 90%)

$$\overline{\mathrm{REP}} = \frac{1}{n_{\mathrm{sim}}} \sum_{i=1}^{n_{\mathrm{sim}}} \sum_{j=1}^{n} \frac{\mathbb{1}(\text{any randomization probability at time } j \ < 10\% \text{ or } > 90\%)}{n}$$

  with indicator function $\mathbb{1}(\cdot)$.

- The proportion of simulations where the number of allocations to treatment 1 was at least 10% of the total sample size n less than the average sample size in the remaining groups

$$\hat{S}_{0.1} = \frac{1}{n_{\mathrm{sim}}} \sum_{i=1}^{n_{\mathrm{sim}}} \mathbb{1}\left(\frac{n - n_{1i}}{K} - n_{1i} > 0.1n\right)$$

  where n_{1i} is the number of allocations to treatment group 1 in simulation i. For $K = 1$, this reduces to the $\hat{S}_{0.1}$ sample size imbalance measure from Robertson et al. (2023), which in turn was inspired by the performance evaluation in the simulation study from Thall et al. (2015). Note that the $\hat{S}_{0.1}$ measure is difficult to translate into an actual patient benefit measure because the patient benefit loss related to imbalance depends on the size of the treatment effect (Robertson et al., 2023). For instance, imbalance is less consequential if the treatment effect is small than if it is large. Nevertheless, we include the $\hat{S}_{0.1}$ measure for comparability with previous simulation studies.

Parameter estimation and hypothesis testing performance was quantified with:

- The empirical bias of the estimate of the rate difference RD_1

$$\text{Bias}(\text{RD}_1) = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \hat{\theta}_{1i} - \hat{\theta}_{0i} - \text{RD}_1$$

  where $\hat{\theta}_{1i}$ and $\hat{\theta}_{Ci}$ are the maximum likelihood estimates of the probabilities θ_1 and θ_C in simulation repetition i.

- Empirical coverage of the 95% Wald confidence intervals of RD_1

$$\text{Coverage}(\text{RD}_1) = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \mathbb{1}\left\{95\% \text{ CI includes } \text{RD}_1 \text{ in simulation } i\right\}.$$

- Empirical rejection rate (type I error rate or power depending on the condition) related to the Wald test of the rate difference RD_1

$$\text{RR}(\text{RD}_1) = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \mathbb{1}\left\{\text{Test rejects } H_0\colon \text{RD}_1 = 0 \text{ in simulation } i\right\}.$$

Each condition was simulated 10'000 times. This ensures a MCSE (Monte Carlo Standard Error) for type I error rate, power, and coverage of at most 0.5%. MCSEs were calculated using the formulae from Siepe et al. (2024) and are provided for all measures in the following figures and the supplemental website.

B.6 Computational aspects

The simulation study was run on a server running Debian GNU/Linux forky/sid and R version 4.5.2 (2025-10-31). The SimDesign R package was used to organize and run the simulation study (Chalmers and Adkins, 2020). Our newly developed `brar` R package was used to perform Bayesian RAR. Code and data to reproduce this simulation study are available at `https://github.com/SamCH93/brar` and persistently archived at `https://doi.org/10.5281/zenodo.17248628`.

B.7 Results

Convergence Non-convergence happened only rarely for the normal approximation method due to logistic regression not converging at the start of the study when no events were observed for some groups. In this case, equal randomization was applied. The highest rate of such non-convergence was in a condition with $K = 3$ treatment groups where 4.6% of the n logistic regressions did not converge. No other forms of missingness were observed. Figures and tables with per-condition-method non-convergence rates are available at https://samch93.github.io/brar/.

Rate of successes and extreme randomization probabilities The mean rate of successes is shown in Figure 10. It was generally the highest for the Gittins index, followed by Bayes UCB, Thompson sampling, and being the lowest for equal randomization. The normal approximation and exact method produced mostly similar rates, with the normal method sometimes showing slightly higher rates (e.g., for $K = 3$ and $\mathrm{RD}_1 = 0.2$). The Thompson sampling modifications generally reduced the mean success rate, with the greatest reduction achieved by combining capping and power transformation with $c = i/(2n)$. In conditions with small sample size these rates were similar as when the prior probability was $H_0 = 0.75$, while they were lower when the sample size was larger. This makes sense as for larger sample sizes, uncapped randomization probabilities are more likely to converge to extreme ones. This is also visible in Figure 11 where more extreme randomization probabilities are observed with increasing sample size and rate difference for all methods but the ones with capping. Finally, Bayes UCB and the Gittins index methods always produced extreme randomization probabilities, as these methods deterministically allocate the next patient to the group with the highest upper confidence bound for and Gittins index, respectively.

Negative sample size imbalance Figure 12 shows negative sample size imbalance as quantified by the $\hat{S}_{0.1}$ metric. The Gittins index generally showed the highest negative imbalance, followed by Thompson sampling and Bayes UCB. Among the methods with $\Pr(H_0) < 1$, negative imbalance was the greatest for Thompson sampling and reduced when

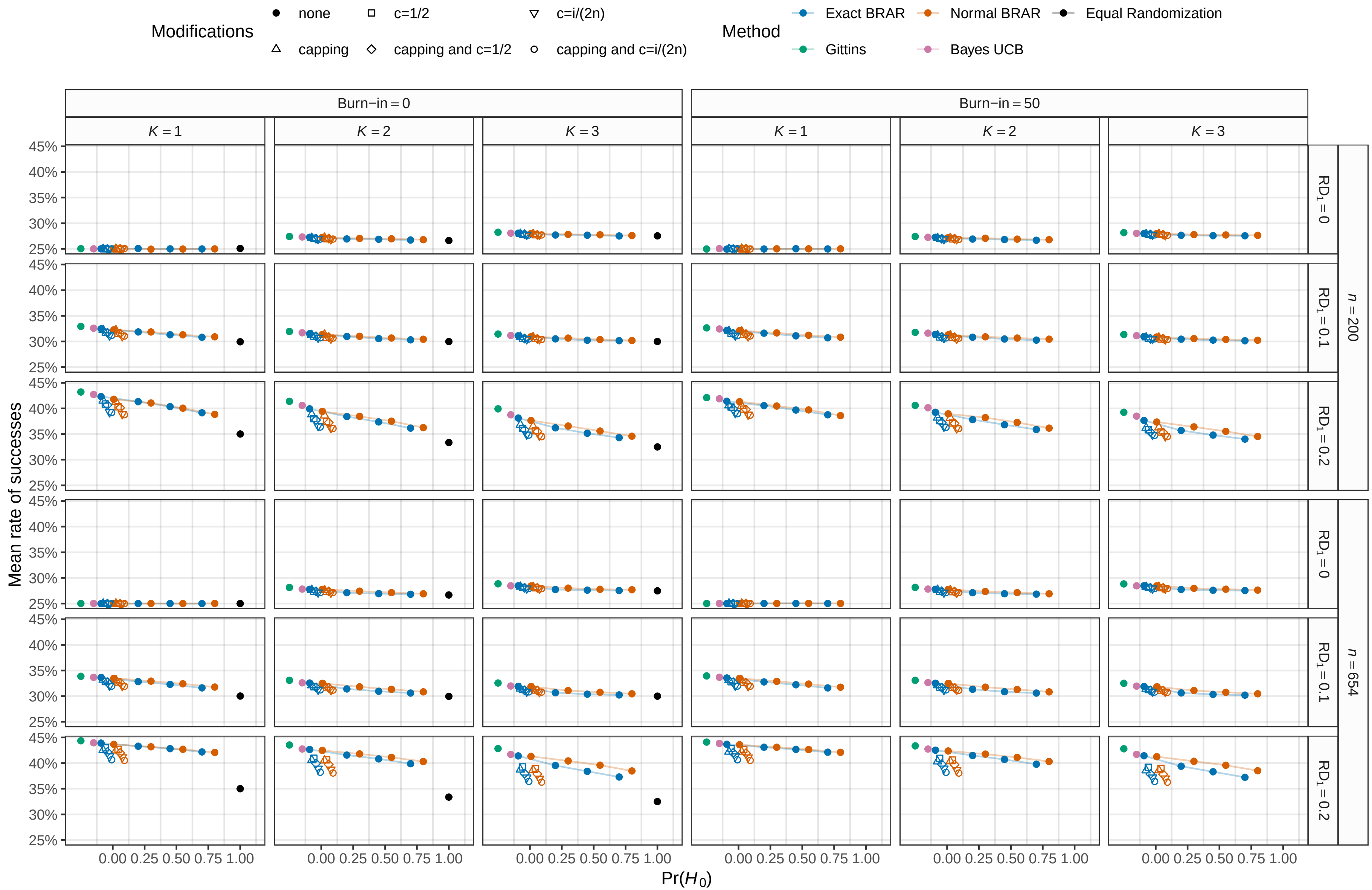

Figure 10: Mean rate of successes (i.e., the number of successes in a study divided by its sample size averaged over all 10’000 simulation repetitions). The maximum MCSE is 0.051%.

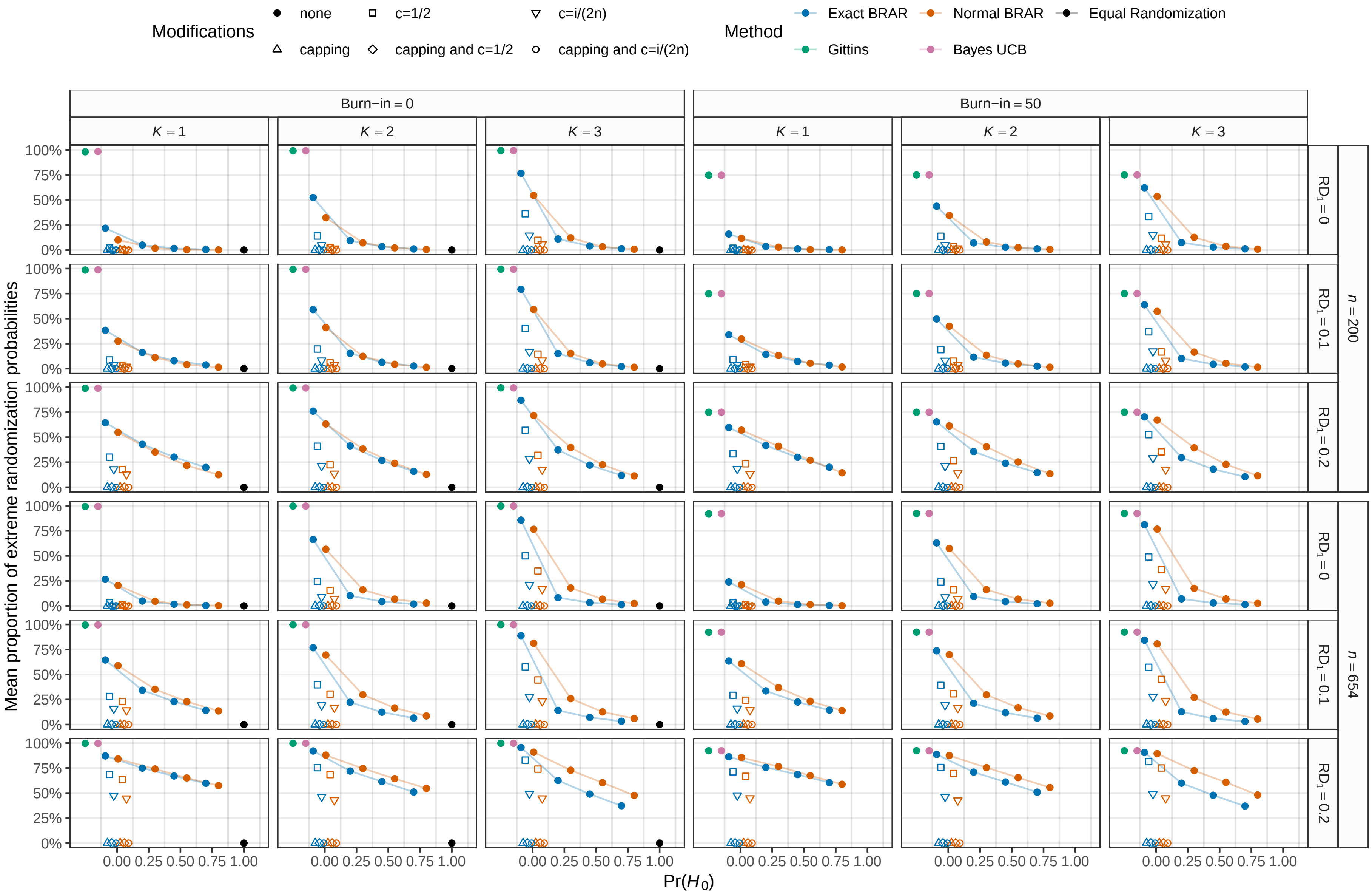

Figure 11: Mean proportion of 10′000 simulations with randomization probabilities either less than 10% or greater than 90%. The maximum MCSE is 0.33%.

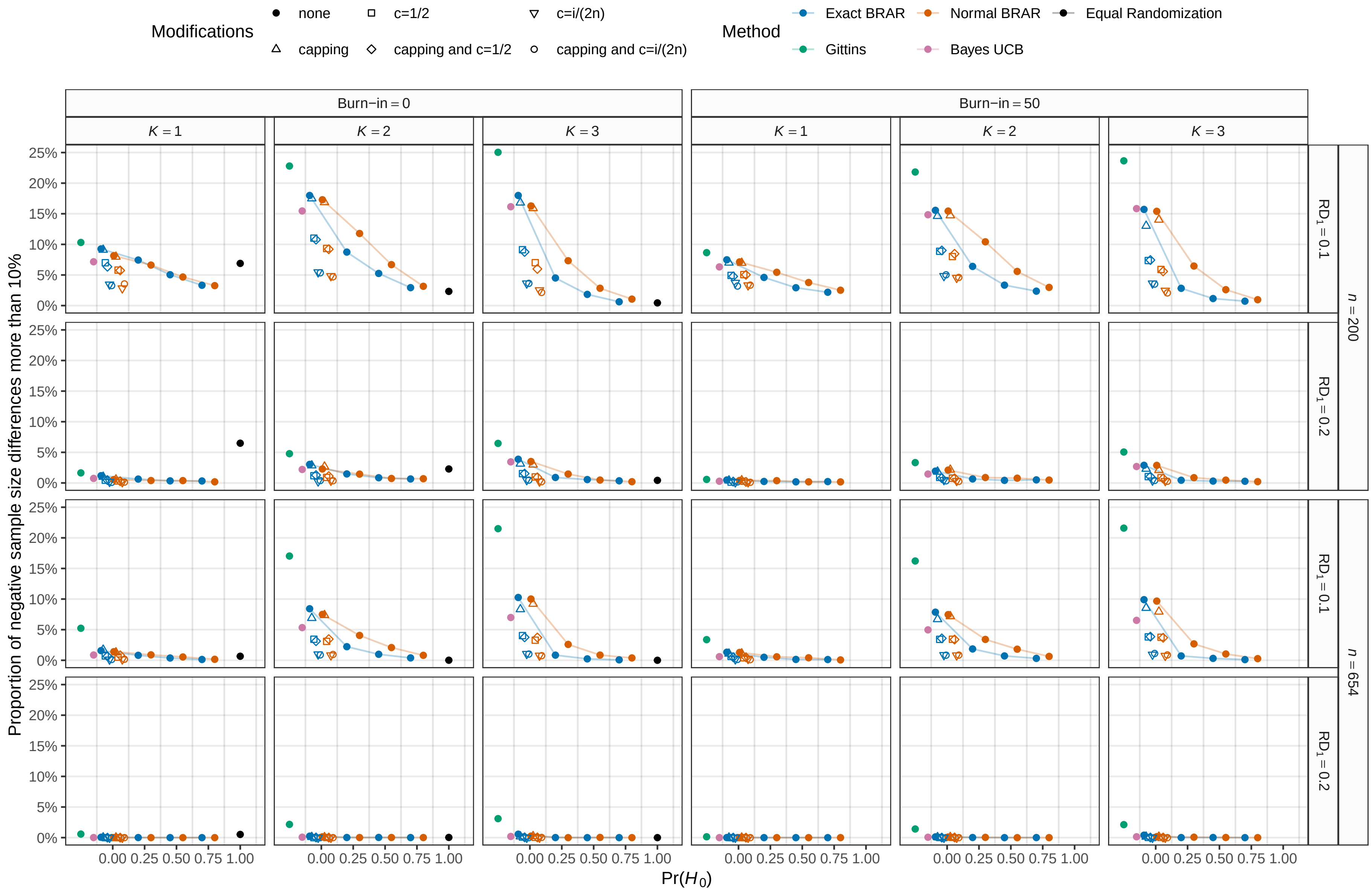

Figure 12: Proportion of 10′000 simulations with more than 10% of the sample size randomized to other groups than treatment group 1. The maximum MCSE is 0.5%.

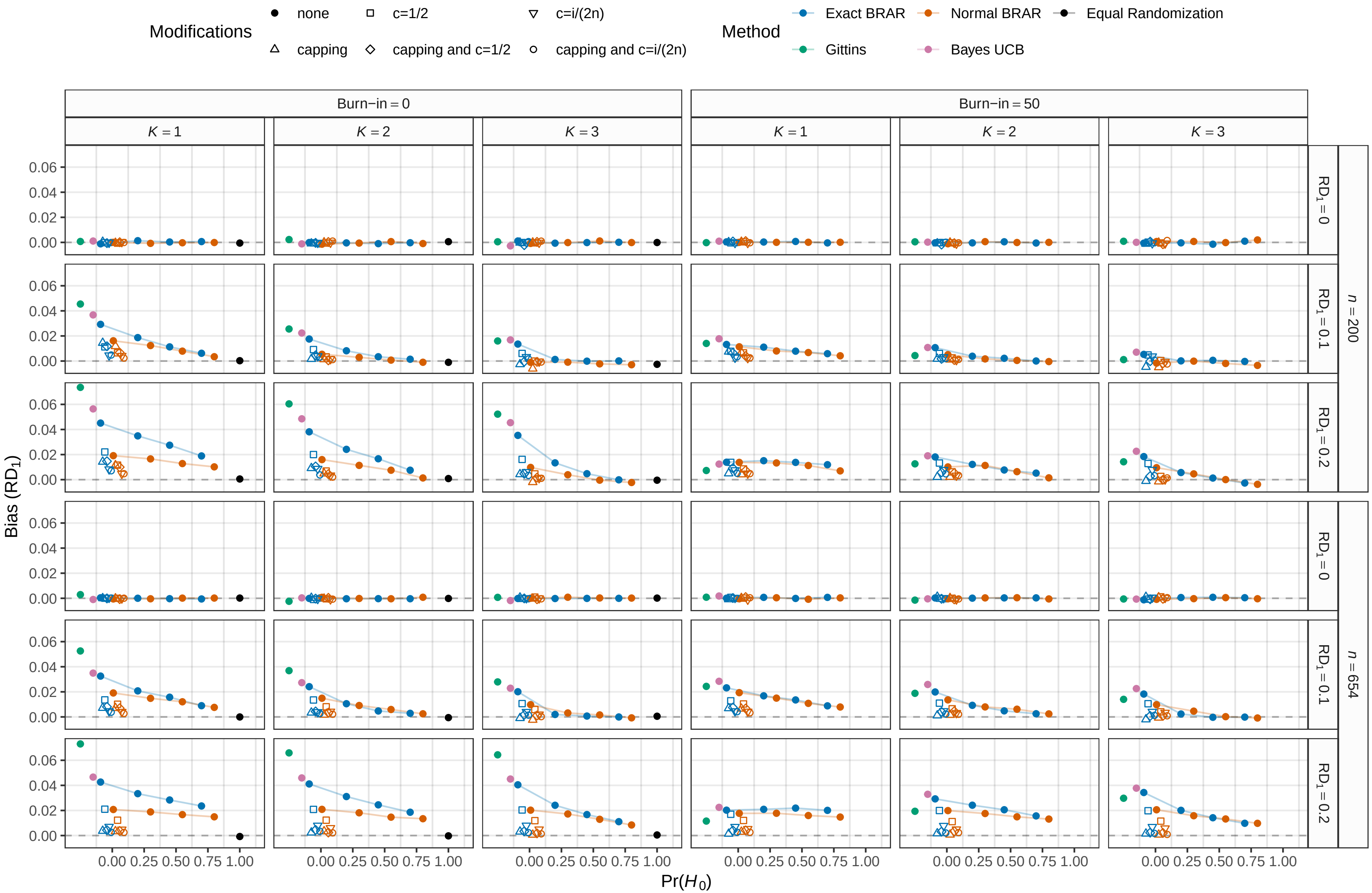

Figure 13: Empirical bias of the estimate of the rate difference RD_1 between the first treatment group and the control group based on 10′000 simulation repetitions. The maximum MCSE is 0.0016.

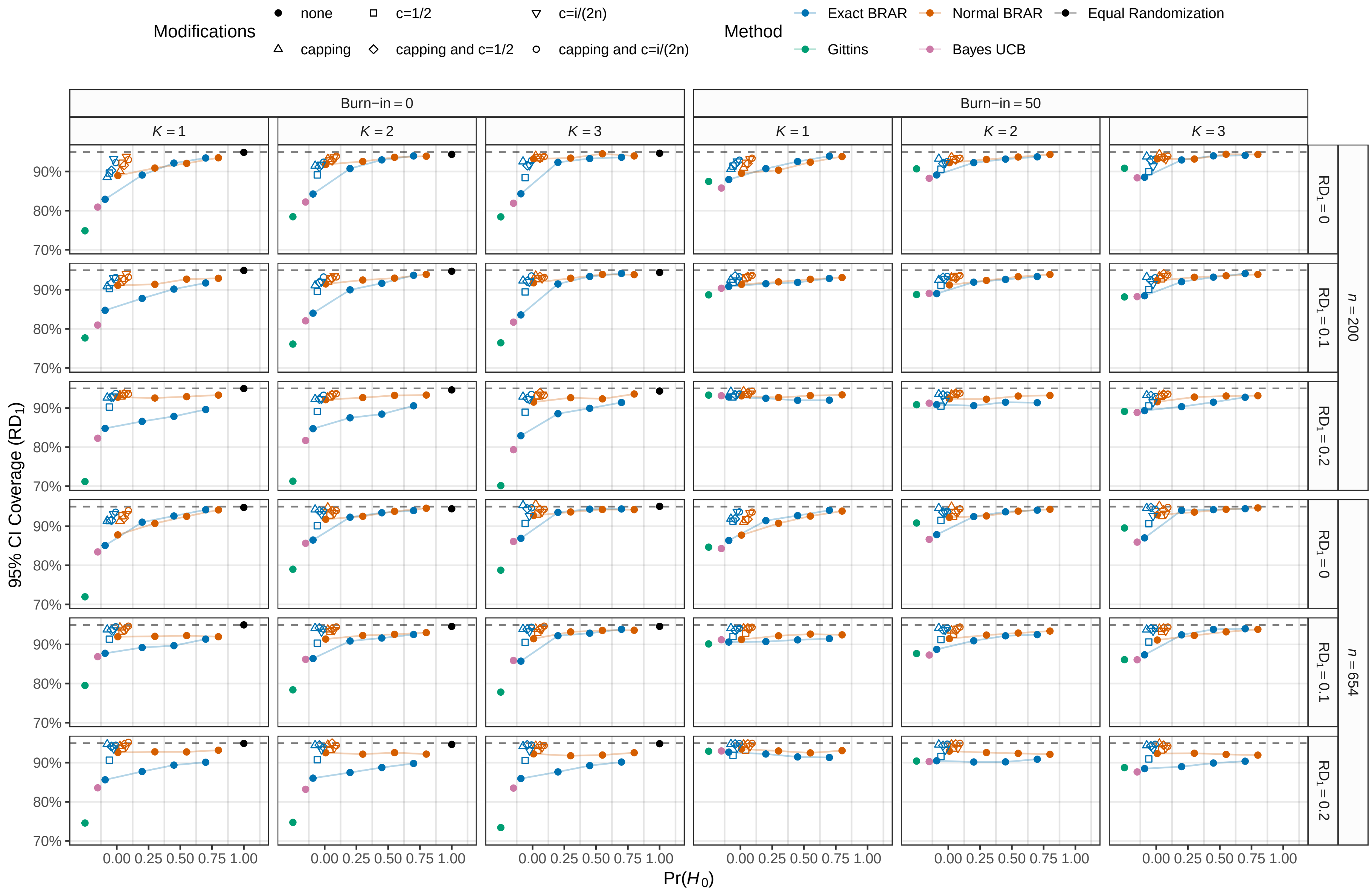

Figure 14: Empirical coverage of the 95% Wald confidence interval for the rate difference RD_1 based on 10′000 simulation repetitions. The maximum MCSE is 0.46%.

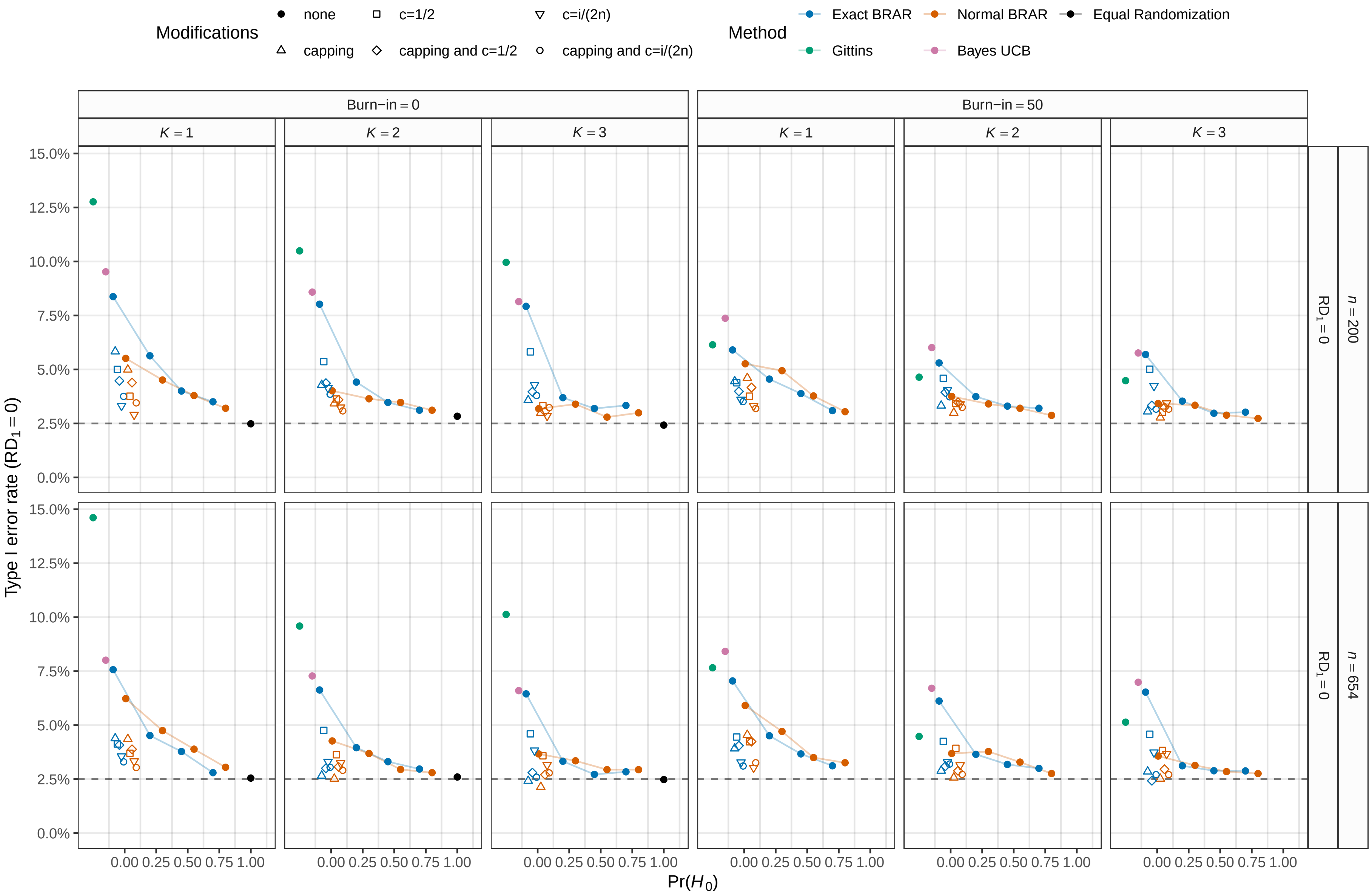

Figure 15: Empirical type I error rate of the Wald test of $\mathrm{RD}_1 = 0$ based on 10′000 simulation repetitions. The maximum MCSE is 0.35%.

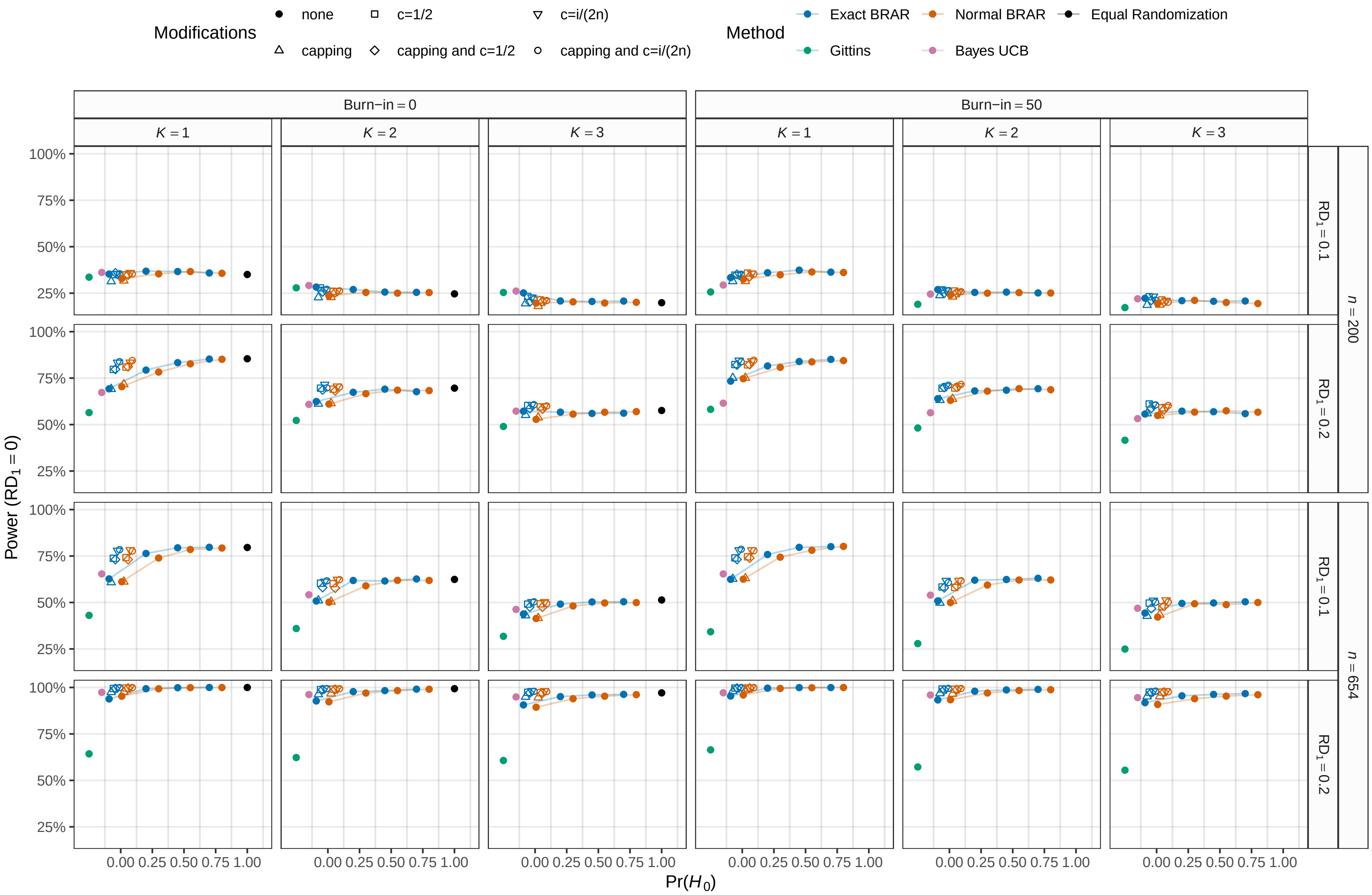

Figure 16: Empirical power of the Wald test of $\mathrm{RD}_1 = 0$ based on 10′000 simulation repetitions. The maximum MCSE is 0.5%.

modifications were applied. Similarly, increasing the prior probability of H_0 decreased negative imbalance, in some cases even below Thompson sampling with modifications. As in the simulation study from Robertson et al. (2023), negative imbalance of Thompson sampling and Bayesian RAR with $\Pr(H_0) < 1$ was higher than under equal randomization in conditions with small effect size $\text{RD}_1 = 0.1$, but equal or lower than under equal randomization in conditions with large effect size $\text{RD}_1 = 0.2$.

Bias and coverage Figures 13 and 14 show empirical bias and coverage related to estimates of the rate difference between the first treatment group and the control group RD_1. In conditions where there was no difference ($\text{RD}_1 = 0$), all methods produced unbiased point estimates, though all methods but equal randomization also showed undercoverage. Bias occurred in conditions with non-zero rate differences, the bias being the greatest for the Gittins index, Bayes UCB, and Thompson sampling, and decreasing to some extent when modifications were introduced or the prior probability of H_0 increased. Similarly, modifications or increasing prior probabilitities improved coverage, although coverage still remained suboptimal in most conditions. For large rate difference conditions, coverage was often better for the normal than the exact version of RAR. Similarly, bias was in some cases somewhat larger for the exact compared to the normal version. This could potentially be due to the normal prior distribution on the log odds ratio being better tuned to effect sizes in the data-generating mechanism difference compared to the uniform priors on the probabilities under the binomial model.

Type I error rate and power Figures 15 and 16 show empirical type I error rate and power associated with the Wald test of $\text{RD}_1 = 0$. We see that the Gittins index, Bayes UCB, and Thompson sampling methods showed an inflated type I error rate above the nominal 2.5% which was reduced to some extent by increasing either the prior probability of H_0 or applying modifications. In the same way, Gittins index, Bayes UCB, and Thompson sampling often exhibited reduced power compared to equal randomization, which was again alleviated by modifications or increasing $\Pr(H_0)$. In small sample sizes and large rate differences, power was slightly increased for the exact compared to the normal version of RAR, while for Thompson sampling the type I rate was slightly higher for the exact compared to the normal version.

Computational details

```r
cat(paste(Sys.time(), Sys.timezone(), "\n"))
```

```
## 2026-02-23 10:43:21.749525 Europe/Zurich
```

```r
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 24.04.4 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0  LAPACK version 3.12.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=de_CH.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=de_CH.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=de_CH.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=de_CH.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Zurich
## tzcode source: system (glibc)
##
## attached base packages:
## [1] parallel  stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
## [1] Ternary_2.3.5  ggpubr_0.6.2   ggh4x_0.3.1    ggplot2_4.0.2  mvtnorm_1.3-3
## [6] brar_0.1       SimDesign_2.21 dplyr_1.2.0    knitr_1.50
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6        xfun_0.54           PlotTools_0.3.1
##  [4] rstatix_0.7.3       lattice_0.22-7      vctrs_0.7.1
##  [7] tools_4.5.2         generics_0.1.4      tibble_3.3.1
## [10] highr_0.11          pkgconfig_2.0.3     R.oo_1.27.1
## [13] RColorBrewer_1.1-3  S7_0.2.1            lifecycle_1.0.5
## [16] compiler_4.5.2      farver_2.1.2        brio_1.1.5
## [19] codetools_0.2-20    carData_3.0-5       httpuv_1.6.16
## [22] htmltools_0.5.9     Formula_1.2-5       later_1.4.4
## [25] pillar_1.11.1       car_3.1-3           tidyr_1.3.2
## [28] R.utils_2.13.0      sessioninfo_1.2.3   abind_1.4-8
## [31] mime_0.13           parallelly_1.46.1   tidyselect_1.2.1
```

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## [34] digest_0.6.39        future_1.69.0        purrr_1.2.1
## [37] listenv_0.10.0       labeling_0.4.3       cowplot_1.2.0
## [40] fastmap_1.2.0        grid_4.5.2           cli_3.6.5
## [43] beepr_2.0            magrittr_2.0.4       broom_1.0.10
## [46] future.apply_1.20.1 clipr_0.8.0          withr_3.0.2
## [49] promises_1.5.0       scales_1.4.0         backports_1.5.0
## [52] sp_2.2-0             globals_0.18.0       audio_0.1-11
## [55] otel_0.2.0           gridExtra_2.3        progressr_0.18.0
## [58] ggsignif_0.6.4       R.methodsS3_1.8.2    shiny_1.12.1
## [61] pbapply_1.7-4        evaluate_1.0.5       RcppHungarian_0.3
## [64] testthat_3.3.0       viridisLite_0.4.3    rlang_1.1.7
## [67] Rcpp_1.1.1           xtable_1.8-4         glue_1.8.0
## [70] R6_2.6.1
```